

Computational Support for a Human-Centered Methodology that Aims to Acquire Database Keys with Perfect Precision and Recall

Extended Version

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Abstract

Database constraints are of utmost importance for the correct modeling, high quality, and efficient operation of databases. Keys, in particular, are fundamental to entity integrity and provide sound and efficient access to data. The keys that should be declared are those that hold on all valid instances, since they express rules of the application domain. Without domain knowledge, some keys are incorrectly perceived as rules, which decreases precision, and some rules are not recognized, which decreases recall. We propose a human-centered methodology whose aim is to close the gap towards perfect precision and recall by providing computational support to domain experts. For keys, we develop two families of algorithms that generate Boolean questions for experts and guarantee that no candidate is overlooked, so that the acquired keys are exactly those determined by the experts' answers. We show that enumerating all questions is as hard as the transversal hypergraph problem, and that our dualization-based family asks a number of questions linear in the number of minimal keys and maximal antikeys. Experiments show which strategies suit which distributions of keys, with the dualization-based family saving up to 99.8% of the questions on real-world data; that the F1 measure of keys mined from data leaves a gap of 48.7 to 97.5% that interviews address; that in an interview study with ten participants of three levels of job expertise, expertise rather than the traversal strategy determines how much of the gap is closed, and a team of mixed expertise can reach the quality of a domain expert; and that focusing interviews on prime attributes makes them feasible on large schemata.

CCS Concepts

• **Human-centered computing** → *Interaction design*; • **Information systems** → **Relational database model**; **Integrity checking**; **Inconsistent data**; **Database utilities and tools**.

Keywords

Algorithm, Database key, Domain expert, Experiment, Interview, Large language model, Requirements acquisition, User study

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1 Introduction

Database constraints form a declarative approach towards ensuring that the semantics of application domains is represented adequately within database systems [12, 18, 51]. In fact, constraints restrict database instances to those that satisfy all rules of the application domain. As example, just as no one in the real world should have multiple valid passport numbers of the same country, this rule should be declared, monitored, and enforced within database systems that model this information. Without sound representation, the two most essential database tasks are compromised: update operations may lead to violations of data integrity, and any analytical method applied to invalid data will not produce insight. Hence, identifying those database constraints that need to be declared and enforced on a database schema is of utmost importance

While over one hundred different classes of database constraints have emerged [13, 53], the perhaps most important classes are domain constraints, keys and foreign keys. These are the only classes that enjoy native support in database systems to ensure domain integrity, entity integrity, and referential integrity, respectively [12, 13]. We focus on keys. A key is a finite set K of columns such that no two different records in a table can have values matching on all columns of K [40]. Due to their role in implementing entity integrity, keys are essential for all data models, including relational [40], conceptual [54], object-relational [32], XML [10], RDF [38], temporal [59], spatial [25], probabilistic [9], and graph [2] models. They are fundamental to data modeling, database design, indexing, and query optimization, as they let us reference entities uniquely, minimize data redundancy, improve selectivity estimates, and offer new access paths and physical designs. They also facilitate data integration [11], the detection of duplicates and anomalies [45], data repair and cleaning [27], and the reliable use of large language models and AI agents for data processing [39]. The discovery of keys from data is one of the core activities in data profiling [1].

In this work we address the problem of identifying the set of meaningful keys for a given schema. This problem is challenging for several reasons. Firstly, it is more general than the well-researched problem of mining the minimal keys from data [1]. Data may not be available or not reliable enough, and the output of key mining algorithms consists of the minimal keys that hold, exactly or approximately, on a given data set. These typically include many meaningless keys and miss some meaningful ones. Secondly, domain knowledge must be entwined in the acquisition process, in the form of human or artificial experts (LLMs) whose opinions may differ. Thirdly, for a schema with n columns there are 2^n candidate keys, and even the set of minimal keys can contain up to $\binom{n}{\lfloor n/2 \rfloor}$ elements. Finally, the problem is very real. Even for schemata curated

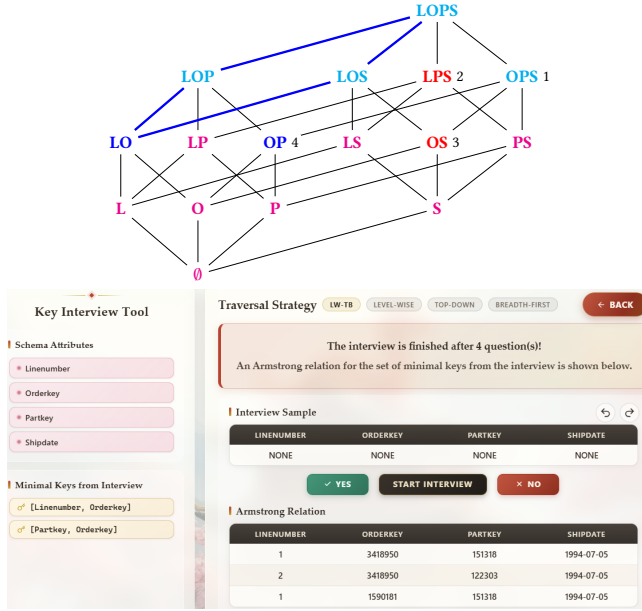


Figure 1: Top-down breadth-first traversal of the level-wise family with primary key $\{L, O\}$ as input for interviewing the set of minimal keys, and final result of tool shown with an Armstrong database

for decades, the meaningful keys may only be known partially. In the HOCKEY database¹, for example, nine out of 22 tables do not have a primary key, and many reasonable UNIQUE constraints have not been specified.

The goal that guides our work is to enable domain experts to determine the set of meaningful keys with perfect precision and recall, and with as little effort as possible. Perfect precision means that no meaningless key is specified, since such a key would prevent valid data from entering the database. Perfect recall means that every meaningful key is specified, since a missing key would allow invalid data to enter the database. We stress that perfect precision and recall is the aim towards which our methodology closes the gap, not a property we claim for its outcome. What our algorithms do guarantee is that no candidate is overlooked: every column set is either asked about or settled by earlier answers, so the acquired keys are exactly those that the experts' answers determine. How close they come to the actual rules depends on the expertise brought to the interview, which our experiments quantify.

Example 1.1. As preview to our contribution consider the largest table of the TPC-H benchmark. Domain knowledge in form of a human expert or LLM suggests that meaningful keys may only be made up of the following four columns: L (inumber), O (rderkey), P (artkey), and S (hipdate). The primary key of the table is $\{L, O\}$, but $\{O, P\}$ constitutes another meaningful minimal key. Fig. 1 shows the underlying Boolean lattice of column sets that needs traversal to identify these meaningful minimal keys, with the latter shown in dark blue. Super-keys are marked in light blue, while anti-keys

¹<https://relational.fel.cvut.cz/dataset/Hockey>

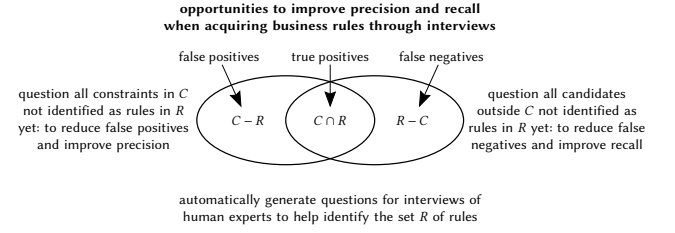


Figure 2: Opportunities to improve the precision and recall when acquiring meaningful database keys

(column sets that are not keys) are marked in magenta and maximal anti-keys in red. The numbers next to the column sets indicate the order in which they are traversed, which in this case follows a top-down, breadth-first strategy. Here, the primary key is already input as a predetermined minimal key to the interview process. Hence, none of its super-keys need traversing, and the interview finishes after four questions instead of the eight that this strategy needs when no minimal key is predetermined. The bottom figure shows the final result with the set of meaningful minimal keys fully identified, together with a database instance that satisfies all keys and violates all anti-keys, known also as an Armstrong instance. $\square$

We propose a computational methodology that promotes domain experts to first-class contributors in acquiring the meaningful keys of a database schema. Fig. 2 illustrates the gap it aims to close: R denotes the target set of meaningful keys, and C the set of keys currently perceived as meaningful by the team of experts. Our algorithms interactively generate Boolean questions. A *No* means that the key in question is meaningful, while a *Yes* means it is not. Different traversal strategies put the questions in different order, which suits different distributions of meaningful keys across the lattice of column sets, and heuristics keep the process practical despite the worst-case exponential solution space. To the best of our knowledge, ours is the first human-centered methodology for database keys in which human input replaces data as the source of truth, whereas related approaches use human input to validate or repair what data suggests, or acquire constraints of constraint satisfaction problems rather than database keys (Sec. 2). In our methodology, data-driven discovery suggests which keys hold on a given instance, while the interviews decide which keys are rules, and data, where available, only illustrates the questions. Our **contributions** are as follows.

(1) Computationally, we propose algorithms that generate interview questions on database keys automatically and exhaustively, so that, unlike in the manually prompted acquisition of keys [28], no candidate key is overlooked. They form two families of algorithms that generate interview questions, organized along a starting point (bottom or top) and a direction (breadth-first or depth-first). The level-wise family traverses the column lattice in the four resulting ways, and the dualization-based family adapts the Dualize and Advance algorithm [24] from membership queries on a database instance to questions answered by experts, from a single traversal to both starting points and both directions, and to predetermined keys. Complexity results set the expectations: enumerating the interview

questions is as hard as enumerating the edges of a transversal hypergraph, so it is open whether an output-polynomial algorithm exists. Nevertheless, the dualization-based family asks a number of questions linear in the number of minimal keys and maximal anti-keys.

(2) Conceptually, our algorithms form the core of a new methodology for acquiring the meaningful keys of a schema that makes domain experts the decision makers, combining question-driven interviews at schema level with the inspection of Armstrong databases at instance level. Utilizing expert knowledge, we also introduce the heuristic of focusing interviews on prime attributes, which makes the process efficient while aiming to maintain quality, despite the intractability results.

(3) Experimentally, we make our algorithms publicly available as a tool for acquiring database keys [3]. Firstly, we show which algorithms perform well for which distributions of minimal keys, and that the dualization-based family reduces the number of questions on real-world data sets by up to 99.8% over the best level-wise strategy. Secondly, we show that state-of-the-art mining of exact and approximate keys leaves a gap of 48.7 to 97.5% in F1 to the meaningful keys, which highlights the danger of using mining algorithms for decision making. Thirdly, an interview study designed to give first insight, with ten participants grouped by job expertise and experience into domain experts, data practitioners, and non-specialists, shows that, once the relevant attributes are known, the three groups acquire the meaningful keys with an F1 of up to 0.91, 0.59 and 0.36, that expertise rather than the choice of strategy determines how much of the gap is closed, and that a team of mixed expertise reaches the quality of a single domain expert once a moderate majority accepts a key. Finally, we show that focusing interviews on prime attributes, predicted by LLMs, reduces the number of questions by 84% while it raises the quality of LLM interviewees, and turns full-schema interviews that humans cannot complete within half an hour into interviews of a few minutes.

Organization. Sec. 2 discusses related work, Sec. 3 our methodology, and Sec. 4 the complexity results that set our expectations. Sec. 5 describes our algorithms, Sec. 6 our experiments, and Sec. 7 concludes. Proofs and further experiments are in the appendix.

2 Related Work

Previous work on acquiring database constraints, and keys in particular, includes data profiling, Armstrong databases, and manually prompted acquisition. None of them puts domain expertise at the center of decision making.

Discovery of Constraints from Data. *Data profiling* [1] computes the constraints of a given class that hold on a given database instance. Deciding whether there is a key of size at most s is NP -complete [40] and $W[2]$ -complete in s [6], the solution space is worst-case exponential [14], and discovering all minimal keys is strongly related to enumerating all minimal transversals of a hypergraph [5, 17, 42]. Counting minimal keys is $\#P$ -hard [24]; our $W[1]$ -hardness result complements this by showing that it is hard to find an anti-key, which is the main problem for generating interview questions. Despite these limits, a broad range of key mining algorithms [7, 21, 30, 43, 55, 57] and tools [47, 56] handle instances

with many rows and columns, using column-efficient [22], row-efficient [29], hybrid [48, 57], approximate [33, 49], incremental [8], and distributed [50] strategies, with a primary focus on efficiency. Recent efforts exclude meaningless constraints during mining [35] or rank the outputs [30, 58]. However, since real data is often inaccurate, incomplete, inconsistent, and outdated, precision and recall of rule discovery from data remain low, as our experiments confirm. Among mining algorithms, Dualize and Advance [24] discovers the minimal keys of an instance through membership queries in time linear in the number of keys and anti-keys it discovers. We adapt it to interviews, in which domain experts replace the instance as the source of truth (Sec. 5.4).

Human-in-the-Loop Data Management. Human input has long guided data management tasks for which data alone is insufficient. In user-guided data repair, users confirm suggested updates ranked by their expected benefit [?], or validate attributes so that editing rules and master data yield certain fixes [?]. Interactive cleaning generalizes single user corrections into cleaning queries [?], KATARA [?] validates table patterns and repairs through knowledge bases and crowdsourcing, and crowdsourced entity resolution lets humans decide on record pairs [?]. Human-guided discovery of constraints has been studied in constraint programming, where constraint acquisition learns a constraint network from a user's answers to membership and partial queries [?], which is closest in spirit to our interviews. However, it targets constraint satisfaction problems rather than database keys, and exploits neither the column lattice, Armstrong databases, nor prime attributes. Foundation models can perform data wrangling tasks [?], which motivates LLMs as interviewees. Unlike these approaches, which use human input to validate or repair what data or rules suggest, our methodology makes human input the source of truth for which keys are rules.

Armstrong Databases. An instance is *Armstrong* for a set C of constraints if it satisfies exactly the constraints implied by C [4, 20]. Such perfect examples can improve the recall of identifying meaningful constraints [37, 41, 44], since every rule incorrectly perceived as meaningless is violated by the instance, and experts may spot and correct such violations. However, few classes enjoy Armstrong databases [19], and while keys do, the minimum size of an Armstrong relation for minimal keys over n columns is at least $\frac{1}{n^2} \binom{n}{\lfloor n/2 \rfloor}$ [15], which makes it hard for experts to spot and correct inconsistencies. Our methodology and tool integrate the inspection and modification of Armstrong relations.

Constraint Acquisition Prompted by Experts. To overcome these limitations, previous work [28] generates small samples that satisfy the perceived constraints C and violate only the constraint in question, but leaves it to experts to decide which constraints to question. Our algorithms fully automate this choice and guarantee that all candidates are considered.

3 Methodology

We describe our methodology for acquiring meaningful constraints of a given class, which aims at perfect precision and recall. While it is not limited to a specific class, we focus on keys as proof of concept.

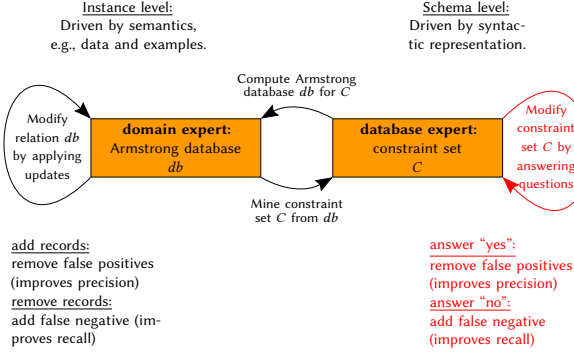


Figure 3: Methodology towards Acquiring the Set of Meaningful Database Keys with Perfect Precision and Recall

Preliminaries. A *table schema* is a finite, non-empty set T of *attributes*, the columns of a table, each with a domain of values. A *record* over T assigns to each attribute a value from its domain, and a *table* over T is a finite set of records over T . A *key* over T is a set $K \subseteq T$, and a table t *satisfies* K iff there are no two different records in t with matching values on all columns of K . A set C of keys *implies* key K iff every table satisfying all keys in C satisfies K , which is the case iff $K' \subseteq K$ for some $K' \in C$. K is a *minimal key* (or *candidate key*) iff no proper subset of K is a key, and an *anti-key* for C iff it is not implied by C . A table is *Armstrong* for C iff it satisfies every key implied by C and violates every other key. Every set of keys enjoys Armstrong tables. The *rules*, or *meaningful keys*, are those satisfied by every table that can occur in the application domain.

Problem and Approach. Given a schema and a possibly empty set C of keys that a team of experts currently perceives as meaningful, the goal is to support the team in revising C such that it coincides with the set R of rules, see Fig. 2. The team may comprise stakeholders such as users, domain and database experts, or LLMs. C may come from earlier stages, from discovery algorithms, from inspecting Armstrong databases, or from prompted acquisition. Fig. 3 shows our methodology, where the novel part is highlighted in red. The tool automatically generates a Boolean question whether the key K in question can be violated, exemplified by a table of two records whose values match exactly on the columns of K . This framing mirrors precisely how a key is violated in data: a key fails exactly when two distinct records duplicate their values on it, so each question presents the smallest possible violation, two records that agree on the columns of K and differ on all others, and asks whether such a duplication can occur in the application domain. Experts thus judge a concrete, minimal counterexample in the vocabulary of their domain, without needing to know what a key is. If the experts answer *No*, K is added to C , otherwise C remains unchanged. Depending on the answer and a traversal strategy, the next question is generated, until all candidates have been considered or the experts stop. No key is ever asked if its status follows from C or from earlier answers: once K is a key, all its supersets are keys, and once K is not a rule, none of its subsets is.

How the Gap is Closed. The methodology offers four ways to close the gap to perfect precision and recall. At schema level,

answering “No” adds a false negative to C , which improves recall, and experts may question a key of C out of order, as in prompted acquisition [28], where answering “Yes” removes a false positive and improves precision. At instance level, driven by the semantics of an Armstrong database db for C , experts may add a record that violates a key, which removes a false positive, or remove a record, which validates a key and removes a false negative. Mining the modified db then switches back to the schema level. The process is deliberately flexible: it may be repeated until the team is satisfied, terminated early, restricted to keys of a given size or to selected attributes, and use real-world records where available. Ultimately, how close the result comes to R depends on the experts’ answers.

4 Complexity: Setting Expectations

Our complexity results are not meant as a main contribution. They set realistic expectations for any interview process [16, 46]. Given a set C of keys over T , the next question of an interview concerns an *anti-key* for C , that is, a column set $A \subseteq T$ such that no $K \in C$ satisfies $K \subseteq A$, and to establish minimality all *maximal* anti-keys need to be interviewed. A column set A is an anti-key for C iff its complement $T \setminus A$ hits every key in C [26]. We study generating questions from a parameterized and an enumeration perspective.

Parameterized complexity. A *parameterized problem* pairs every instance x with a natural number k , the *parameter*. It is *fixed-parameter tractable*, and belongs to the class FPT, if it can be decided in time $O(f(k) \cdot p(|x|))$ for a computable function f and a polynomial p . Above FPT lies the W-hierarchy $\text{FPT} \subseteq \text{W}[1] \subseteq \text{W}[2] \subseteq \dots$, which is not known to be strict. An *fpt-reduction* maps an instance (x, k) in FPT time to an equivalent instance (x', k') with $k' \leq g(k)$ for a computable function g , and $\text{W}[1]$ -hardness under fpt-reductions is the parameterized analogue of NP-hardness: a $\text{W}[1]$ -hard problem is not expected to admit an FPT algorithm. Given a set C of keys over T and a natural number k , the problem ANTI-KEY asks whether there is an anti-key A for C with $|A| = k$. Taking k as parameter captures the natural hope that questions about small column sets are easy to generate.

THEOREM 4.1. ANTI-KEY is $\text{W}[1]$ -hard in the parameter k .

Proof sketch. We reduce the $\text{W}[1]$ -complete problem CLIQUE [16] by an fpt-reduction. Given a graph $G = (V, E)$ and k , let the columns be the vertices, $T = V$, and let every non-edge $\{u, v\} \notin E$ be a two-column key in C . A column set then contains no key of C iff it contains no non-edge, so the anti-keys for C of size k are exactly the cliques of G of size k (App. A). Hence, unless $\text{FPT} = \text{W}[1]$, no algorithm finds an anti-key of a prescribed size k in time $f(k) \cdot p(|T| + |C|)$.

Enumeration complexity. An *enumeration problem* maps every instance x to a finite set $\text{Sol}(x)$ of solutions of size polynomial in $|x|$, which are to be listed without duplicates [31, 52]. Since $\text{Sol}(x)$ may be exponentially large, efficiency is measured in the size of input and output together: an algorithm is *output-polynomial* if it runs in time polynomial in $|x| + \sum_{y \in \text{Sol}(x)} |y|$. For a hypergraph $\mathcal{H} = (V, E)$, a *transversal* is a set $X \subseteq V$ with $X \cap e \neq \emptyset$ for all $e \in E$, and the *transversal hypergraph* $\text{Tr}(\mathcal{H})$ consists of the inclusion-minimal transversals. We consider three enumeration problems.

Problem	Input	Output
ANTI-KEY-ENUM	keys C over T	maximal anti-keys for C
ARMSTRONG-ENUM	keys C over T	records of a table Armstrong for C
HYPERGRAPH-TRANSVERSAL-ENUM	hypergraph $\mathcal{H}$	edges of $\text{Tr}(\mathcal{H})$

The last problem is the yardstick of enumeration complexity [6, 23]. Whether it admits an output-polynomial algorithm is a major open question, while the best known algorithm runs in quasi-polynomial time in the size of input and output [?].

THEOREM 4.2. *The three enumeration problems above are pairwise equivalent under polynomial-time reductions that preserve the solutions one-to-one.*

Proof sketch. The theorem collects correspondences known from the study of keys, hypergraph transversals, and Armstrong relations [17, 23, 42], and states them for the interview setting. For the hypergraph $\mathcal{H}_C = (T, C)$, a column set A is a maximal anti-key for C iff $T \setminus A$ is a minimal transversal of $\mathcal{H}_C$, so complementation is a bijection between the solutions of the first and the third problem. For the second problem, fix a record r_0 and add, for every maximal anti-key A , a record r_A that agrees with r_0 exactly on the columns of A and has fresh values elsewhere. Every two records agree on a subset of some maximal anti-key, so the table satisfies every key implied by C , and the pair r_0, r_A violates every subset of A , so the table is Armstrong for C and its records other than r_0 correspond one-to-one to the maximal anti-keys [23] (App. A).

It is therefore open whether the maximal anti-keys for the keys currently perceived as meaningful, or the records of an Armstrong table, can be enumerated in output-polynomial time, and the dualization step of Sec. 5.4 inherits the quasi-polynomial bound. Generating the next question means computing the next edge of a transversal hypergraph, and the interview is a natural practical fit for this: the hard step is only ever needed one edge at a time, between two answers and at the pace of human deliberation, and every answer prunes the hypergraph before the next edge is computed. Rather than worst-case efficiency, we thus aim for algorithms whose number of questions is bounded in the size of their output (Sec. 5.4), and for heuristics that keep interviews practical (Sec. 6.5).

5 Algorithms

In this section, we will present our algorithmic contributions, starting with some overall outline and motivating example, and a description of the algorithms itself.

5.1 Overall Outline

We will present two suites of algorithms that implement fundamental strategies for traversing the Boolean powerset lattice of the underlying schema. Input to every algorithm is the table schema itself, a set of predetermined minimal keys resulting from previous analysis that is possibly empty, and a traversal strategy. The latter has a starting point, which is either the empty set or the entire column set determining whether it is a bottom-up or top-down approach, and a direction, which is either breadth-first or depth-first. The strategy determines the order in which column sets are put forward to a human expert team in form of a Yes/No question. “Yes” means that column set is an anti-key, while “No” means it is a key. The answer determines how the search space is pruned: In the first

case, all subsets will be anti-keys needing no further questioning, and in the second case, all super-sets will be keys requiring no further questioning. Hence, the interview process with the human expert team is very much interactive.

All strategies share the same interview loop and differ only in how the questions of the next round are generated from the answers received so far. We therefore organize our suites as a single framework with three parameters: the two dimensions above, and the *family* of the strategy. The *level-wise* (LW) family moves through the lattice one column at a time, and its four combinations of a starting point and a direction are the top-down breadth-first (LW-TB), the top-down depth-first (LW-TD), the bottom-up breadth-first (LW-BB), and the bottom-up depth-first (LW-BD). The *dualization-based* family, called *dualize-and-advance* (DA), adapts the Dualize and Advance algorithm for discovering all most specific sentences [24] to our interview setting, in which the answers of the domain experts replace the database instance as the source of truth. It jumps directly to those column sets whose status is still open, and instantiates the same two dimensions in the same order, which gives DA-TB, DA-TD, DA-BB, and DA-BD.

Building on Ex. 1.1 we are illustrating all eight traversal strategies in Fig. 4. The numbers next to the column sets indicate the traversal order of candidate keys for which the Boolean questions are presented to the human experts. In this specific case, the distribution of the two minimal keys is best explored by LW-TB in Fig. 4a, which requires eight questions. Fig. 4b illustrates LW-TD that needs 11 questions, Fig. 4c shows LW-BB that needs 12 questions, and Fig. 4d shows LW-BD that needs 11 questions. Fig. 4h shows that DA-BD needs eight questions, matching LW-TB on this example: questions 1–5 advance the empty set to the maximal anti-key *LPS*, confirming the minimal key *LO* along the way; dualization then jumps to the only remaining candidate *O*, whose advancement identifies the second minimal key *OP* and the second maximal anti-key *OS* with questions 6–8. Fig. 4g shows that DA-BB asks the same questions in the same order, since every round of this example offers a single unsettled candidate, which is the only situation in which the two directions cannot differ. The top-down members start from the entire column set and shrink it to a minimal key instead, which takes DA-TB eight questions in Fig. 4e and DA-TD nine in Fig. 4f, the only member of the family that does not match LW-TB on this example.

Based on Sec. 4, the enumeration process is worst-case exponential in the schema size, and it is an open problem whether an algorithm exists that is polynomial in the size of the output. Intuitively, this is the price for being assured that all candidates for minimal keys have been considered. However, the dualize and advance family operates in time linear in the number of minimal keys and maximal anti-keys. Ultimately, however, which algorithm performs best depends largely on the distribution of the output. Experiments will investigate this question and others in Sec. 6.

5.2 The Interview Framework

Alg. 1 forms the basis for instantiating all eight traversal strategies. It maintains the set R of column sets known to be keys, initialized with the predetermined minimal keys R_{in} , and the set A of column sets known to be anti-keys. Three rules propagate the answers

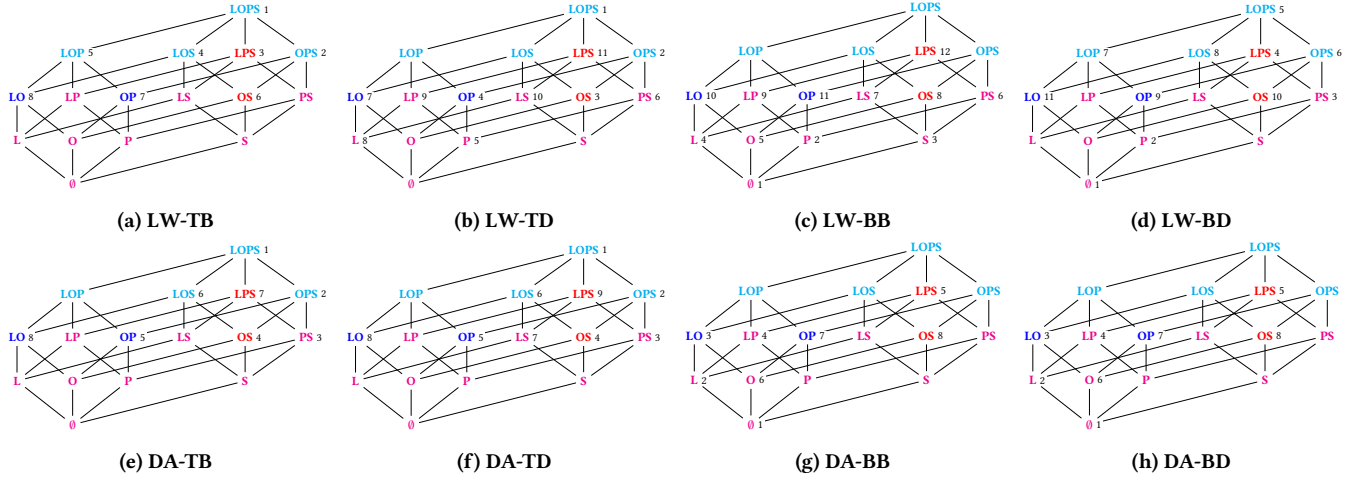


Figure 4: Traversals on running example: level-wise strategies at the top and dualize-and-advance strategies at the bottom

Table 1: Our eight strategies, obtained by instantiating Alg. 1 with a family, a starting point, and a direction

family	bottom-up ($\trianglelefteq$ is $\subseteq$)		top-down ($\trianglelefteq$ is $\supseteq$)	
	breadth	depth	breadth	depth
level-wise (Alg. 2)	LW-BB	LW-BD	LW-TB	LW-TD
dualization (Alg. 3)	DA-BB	DA-BD	DA-TB	DA-TD
questions of the dualization family	$\leq \mathcal{K} + (T + 1) \cdot \mathcal{K}^{-1} $ Thm. 5.1		$\leq \mathcal{K}^{-1} + (T + 1) \cdot \mathcal{K} $ Cor. 5.2	

received so far: i) a column set is a key if it contains some member of R , ii) it is an anti-key if it is contained in some member of A , and iii) it is an anti-key if it is a proper subset of some predetermined minimal key, since the latter is minimal. We call a column set *settled* if one of the three rules applies to it, and write $\text{status}(X)$ for the function in lines 10–15 of Alg. 1, which returns the status of X and asks the experts if and only if X is not settled. Every question of every strategy in our suite is asked in line 13, so the number of interview questions is the number of times that line is executed.

The starting point of a strategy fixes the orientation of the traversal, which lets us describe the bottom-up and the top-down case at once. For the bottom-up starting point, $X \trianglelefteq Y$ denotes $X \subseteq Y$, the successor $X \oplus e$ of X under column e denotes $X \cup \{e\}$, and the *advancing* status is that of an anti-key. For the top-down starting point, $X \trianglelefteq Y$ denotes $X \supseteq Y$, $X \oplus e$ denotes $X \setminus \{e\}$, and the advancing status is that of a key. In both cases, $\trianglelefteq$ is the order in which the traversal proceeds, the $\trianglelefteq$ -least column set is the starting point, and the first two propagation rules say precisely that $X \trianglelefteq Y$ and Y advancing imply that X is advancing. Every round of Alg. 1 therefore collects those candidates that are advancing, and a strategy makes progress by moving from them to their successors. The direction of a strategy decides how many candidates of a round are put to the experts: breadth-first interviews all candidates of the round, while depth-first stops at the first advancing answer and lets the traversal proceed from there, see line 7. The family of a strategy decides how the candidates of the next round are generated from the advancing sets of the current round, see line 9. Tab. 1 summarizes the resulting suite.

Algorithm 1 $\text{keyInterview}(T, R_{\text{in}}, \mathcal{F}, S, D)$

Require: T : table schema, R_{in} : predetermined minimal keys, $\mathcal{F}$: family, S : traversal start, D : traversal direction.

Ensure: Set of minimal keys as perceived by human expert team.

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1:  $R \leftarrow R_{\text{in}}, A \leftarrow \emptyset, N \leftarrow \emptyset$   $\triangleright$  keys, anti-keys, advancing sets
2:  $C \leftarrow \{T\}$  for  $S = \text{top-down}$ , otherwise  $C \leftarrow \{\emptyset\}$ 
3: while true do
4:   for each  $X \in C$  do  $\triangleright$  the interview round
5:     if  $\text{status}(X)$  is advancing then
6:        $N \leftarrow N \cup \{X\}$ 
7:       if  $D = \text{depth-first}$  then break
8:   if  $N = \emptyset$  then return  $\{K \in R \mid K' \not\subseteq K \text{ for all } K' \in R\}$ 
9:    $(C, N) \leftarrow \mathcal{F}(C, N)$   $\triangleright$  Alg. 2 or Alg. 3

10: function  $\text{STATUS}(X)$   $\triangleright$  asks at most one question
11:   if  $r \subseteq X$  for some  $r \in R$  then return key
12:   if  $X \subseteq a$  for some  $a \in A$ , or  $X \subset r$  for some  $r \in R_{\text{in}}$  then
13:     return anti-key
14:   ask the experts whether  $X$  is a key
15:   if the answer is 'No' then  $R \leftarrow R \cup \{X\}$ ; return key
16:    $A \leftarrow A \cup \{X\}$ ; return anti-key

```

5.3 Level-wise Candidate Generation

The level-wise family generates the candidates of the next round as successors of the column sets that are recognized as advancing in the current round, see Alg. 2. Line 1 keeps candidates of the current round that are still unsettled, which are exactly the candidates that a depth-first strategy leaves unasked when it stops at its first advancing answer. Lines 3–7 form the successors of the advancing sets and keep the unsettled ones as candidates. A successor that is settled incurs no question, but if it is advancing, then its own successors still need to be explored, so it is collected in N' and expanded in the next round. If this leaves no candidate at all, then lines 2–8 keep expanding until a candidate is found or no advancing set remains. This is also how the strategies start when predetermined minimal

Algorithm 2 $\mathcal{F}_{\text{lev}}(C, N)$: level-wise generation

Require: C : candidates of round, N : sets recognized as advancing.
Ensure: Candidates C and advancing sets N of the next round.

```

1:  $C \leftarrow \{X \in C \mid X \text{ is not settled}\}$   $\triangleright$  unasked candidates survive
2: repeat
3:    $N' \leftarrow \emptyset$ 
4:   for each  $X \in N$  and each  $e \in T$  with  $X \oplus e \neq X$  do
5:     if  $X \oplus e$  is not settled then  $C \leftarrow C \cup \{X \oplus e\}$ 
6:     else if  $\text{status}(X \oplus e)$  is advancing then  $N' \leftarrow N' \cup \{X \oplus e\}$ 
7:    $N \leftarrow N'$ 
8: until  $C \neq \emptyset$  or  $N = \emptyset$ 
9: return  $(C, N)$ 

```

keys are given, since the traversal then passes through the settled part of the lattice without asking a single question.

Under the convention of Sec. 5.2, Alg. 2 covers all four level-wise strategies. For the bottom-up starting point it extends the anti-keys of the round by one column at a time, and for the top-down starting point it removes one column at a time from the keys of the round. A depth-first strategy interviews the candidates of a round in $\trianglelefteq$ -descending order, so that it continues from the candidate that has advanced furthest, while the order is immaterial for a breadth-first strategy, which interviews all of them. Since status tests every candidate before a question is asked, no strategy of our suite ever puts a column set to the experts whose status is implied by an earlier answer, so every level-wise strategy asks the minimum number of questions its traversal order admits. Their weakness is that this number depends on the size of the levels between the starting point and the minimal keys, rather than on the number of minimal keys.

5.4 Dualization-based Candidate Generation

The dualization-based family avoids this dependence. Compared to the original Dualize and Advance algorithm [24], which poses membership queries to a database instance, our adaptation takes the answers of experts as the source of truth, exploits predetermined minimal keys, and instantiates both starting points and both directions. It maintains the part B of the border that has been identified so far: consisting of maximal anti-keys for the bottom-up starting point and of minimal keys for the top-down starting point; and it only ever asks about column sets that are $\trianglelefteq$ -least among those whose status is still open, see Alg. 3. An advancing answer to such a candidate is a *counterexample*: a column set that is advancing but not $\trianglelefteq$ -below any border element found so far. Lines 2–5 turn each counterexample into a new border element by moving it greedily along the traversal order, one column at a time, and skipping every successor whose status is settled. Line 6 then dualizes: the candidates of the next round are the $\trianglelefteq$ -least column sets that are not $\trianglelefteq$ -below any member of B . For the bottom-up starting point these are the minimal transversals $\text{Tr}(\mathcal{H})$ of the hypergraph $\mathcal{H}$ with vertex set T whose edges are the complements of the maximal anti-keys in B , and for the top-down starting point they are the complements of the minimal transversals of the hypergraph whose edges are the minimal keys in B . For $B = \emptyset$, the $\trianglelefteq$ -least column set is the only candidate, so bottom-up members open with the empty

Algorithm 3 $\mathcal{F}_{\text{dual}}(C, N)$: dualization-based generation

Require: N : counterexamples of the round, B : border found so far, persistent across rounds and initially empty.

Ensure: Candidates of the next round.

```

1: for each  $X \in N$  do
2:   if  $X \trianglelefteq Y$  for some  $Y \in B$  then continue  $\triangleright$  covered
3:   for each  $e \in T$  with  $X \oplus e \neq X$  do  $\triangleright$  advance X
4:     if  $\text{status}(X \oplus e)$  is advancing then  $X \leftarrow X \oplus e$ 
5:      $B \leftarrow B \cup \{X\}$   $\triangleright$  new border element
6: return  $(\{X \subseteq T \mid X \text{ is } \trianglelefteq\text{-least with } X \not\trianglelefteq Y \text{ for all } Y \in B\}, \emptyset)$ 

```

column set like LW-BB and LW-BD, while top-down members open with the full schema T like LW-TB and LW-TD. The candidates of a round are interviewed in $\trianglelefteq$ -ascending order, so that a depth-first member advances the counterexample closest to its starting point first.

The dualization step is what makes the family efficient in the number of questions. Every candidate is $\trianglelefteq$ -least among the column sets whose status is open, so a bottom-up candidate that is confirmed as a key is a minimal key: each of its proper subsets misses some edge of $\mathcal{H}$, is therefore contained in some maximal anti-key of B , and is hence an anti-key itself. Dually, a top-down candidate that is confirmed as an anti-key is a maximal anti-key. The interview of a round thus either confirms border elements on one side, or produces counterexamples that Alg. 3 advances into border elements on the other side. The strategy terminates when a round produces no counterexample, in which case the minimal column sets in R are exactly the meaningful minimal keys, and B contains all maximal anti-keys.

THEOREM 5.1. *The bottom-up members of the DA family acquire the set $\mathcal{K}$ of meaningful minimal keys, as perceived by the experts with respect to their answers, with at most $|\mathcal{K}| + (|T| + 1) \cdot |\mathcal{K}^{-1}|$ questions, where $\mathcal{K}^{-1}$ denotes the set of maximal anti-keys for $\mathcal{K}$.*

COROLLARY 5.2. *The top-down members of the DA family acquire the set $\mathcal{K}$ of meaningful minimal keys, as perceived by the experts with respect to their answers, with at most $|\mathcal{K}^{-1}| + (|T| + 1) \cdot |\mathcal{K}|$ questions.*

Proof sketch. Every question concerns a column set that is not settled. In a bottom-up member, a candidate of the dualization step that is answered ‘No’ is a minimal key, which is never asked again, so at most $|\mathcal{K}|$ questions are answered ‘No’. A candidate answered ‘Yes’ is a counterexample, which the advance step turns into a new maximal anti-key with at most $|T|$ further questions, so at most $|\mathcal{K}^{-1}|$ candidates are answered ‘Yes’. The top-down case is dual (App. A).

Comparing the two bounds, the bottom-up members are preferable when there are fewer maximal anti-keys than minimal keys, and the top-down members are preferable in the opposite case. The direction has no effect on these bounds. A breadth-first member interviews all candidates of a dualization step before their counterexamples are advanced, which reduces the number of dualization steps and allows a team of experts to answer the questions of a round in parallel, at the price of at most one question for each candidate that a counterexample of the same round would have

settled. A depth-first member advances its first counterexample immediately, at the cost of one dualization step per border element.

In contrast to the level-wise family, whose number of questions can be exponential in $|T|$ even when a single minimal key lies in the middle of the lattice, Thm. 5.1 and Cor. 5.2 bound the questions of the DA family linearly in the size of the interview output, that is, the positive and negative border formed by the minimal keys and the maximal anti-keys. This benefit has two drawbacks. Firstly, the dualization step must enumerate the minimal transversals of a hypergraph, which is the yardstick problem of Sec. 4. This computation, however, happens between questions, so it requires computation time of the tool but no effort of the experts, and it remains sub-exponential in the output size [24]. Secondly, the bottom-up members pay up to $|T| + 1$ questions for each maximal anti-key even in cases where a level-wise strategy prunes by use of a single question: on a schema without any key, they advance the empty set through $|T| + 1$ questions to the maximal anti-key T , while LW-TB identifies T as an anti-key with its first question. Within the family, this weakness is removed by the dual starting point: the top-down members need a single question on a schema without any key, and dually the bottom-up members need a single question when the empty column set is a key. Hence, the DA family complements our LW suite: it is attractive whenever few minimal keys and few maximal anti-keys are distributed around the middle levels of the lattice, where the levels are largest and the level-wise strategies ask the most questions.

6 Experiments

We report experiments on four aspects of our methodology: how the strategies of our suite cope with different distributions of minimal keys, how much of a gap key mining leaves, how human interviewees of different expertise benefit from interviews, and how focusing interviews on prime attributes helps. Further experiments, in particular a detailed study of LLMs as domain experts, are in App. B.

6.1 Setup and Research Questions

All non-LLM experiments were implemented in Java 17 and run on a Windows 11 server with a 12th Gen Intel Core i7-12700 CPU @ 2.10 GHz and 128 GB of RAM. The LLM experiments ran on a Red Hat Enterprise Linux 8.10 server with two NVIDIA A100 GPUs (80 GB each), a 256-core CPU, and 1 TB of memory. Source code, the **interview tool**, data sets, and experimental materials [3] are available at <https://anonymous.4open.science/r/interview4key>.

Our research questions are as follows. **RQ1**: How do the two families and their strategies perform under different distributions of minimal keys, and how many questions does the dualization-based family save on real-world data? **RQ2**: How large is the gap between keys mined from data, or declared on a schema, and the meaningful keys? **RQ3**: How much of that gap do human interviewees of different job expertise and experience close with our interviews, individually and as a team? **RQ4**: How much does focusing interviews on prime attributes, that is, attributes that occur in some meaningful minimal key, reduce the effort of an interview, and what does it do to quality?

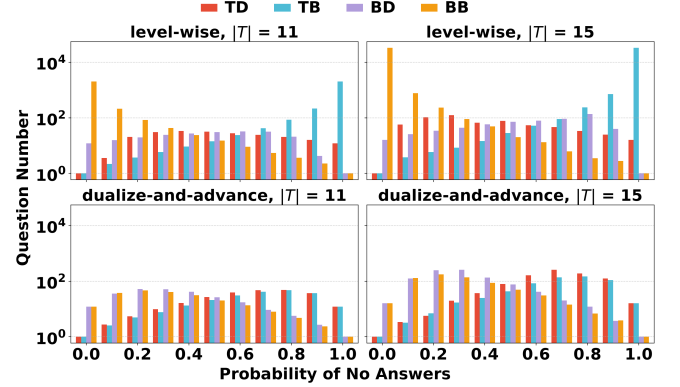


Figure 5: Average number of interview questions of all eight strategies under probabilistic answering, on schemas of $|T| = 11$ and $|T| = 15$ columns. The top row shows the level-wise family and the bottom row the dualize-and-advance family, on a common y axis, so that the bars of the two families can be compared directly. A colour denotes the same traversal in both families.

6.2 Distributions of Minimal Keys

Synthetic distributions. To sweep the whole spectrum of distributions, from no key at all to every column set being a key, we answer ‘No’ with probability p for $p = 0, 0.1, \dots, 1$. Fig. 5 shows the average number of questions of all eight strategies for $|T| = 11$ and $|T| = 15$, with a remarkable symmetry around $p = 0.5$. With no key present ($p = 0$), the top-down strategies identify the full schema as an anti-key with a single question, while LW-BB traverses all levels of the lattice and LW-BD a path from bottom to top. These roles reverse when every column set is a key ($p = 1$). LW-TB is the best performer for $p < 0.5$, since ‘Yes’ answers to large column sets identify maximal anti-keys and prune quickly, and dually LW-BB is best for $p > 0.5$. The two cross between $p = 0.4$ and $p = 0.6$; for instance, at $p = 0.5$ and $|T| = 15$, LW-BB asks 20.3 questions against 28.5 of LW-TB. The depth-first strategies LW-TD and LW-BD are symmetric, too, each being most expensive in the middle of the range and cheap at the extreme that matches its starting point. *Takeaway*: within the level-wise family, the better starting point is the one closer to the border, top-down when keys are rare and bottom-up when they are abundant.

The DA family, in the bottom row of Fig. 5, shows the same symmetry, and its extremes behave as Thm. 5.1 and Cor. 5.2 predict: without keys, the top-down members ask a single question while the bottom-up members advance the empty set to T with $|T| + 1$ questions. What separates the two families is their worst case over p . Every level-wise strategy has a range of p in which it traverses essentially the whole lattice, as LW-BB asks $2^{|T|} = 32,768$ questions at $p = 0$ and LW-TB asks 32,767 at $p = 1$ for $|T| = 15$, while no DA member asks more than 256 questions at any p . The price is that no DA member is best in the middle of the range, for instance 42.6 against 20.3 questions at $p = 0.5$ and $|T| = 15$. Independent random answers induce a large border there, of between 5.0 and 31.4 minimal keys on average, and the questions of a DA member grow

Table 2: Number of interview questions of the 8 strategies on real-world data sets, ordered by schema size. Cells show the total number of questions, and the fewest questions within each family are in bold. $|\mathcal{K}|$ and $|\mathcal{K}^{-1}|$ denote the numbers of minimal keys and maximal anti-keys. The last column is the reduction in the total number of questions of the best dualization-based over the best level-wise strategy.

data set	T	$\mathcal{K} $	$\mathcal{K}^{-1} $	level-wise (LW)				dualization-based (DA)				red. in %
				TD	TB	BD	BB	TD	TB	BD	BB	
abalone	9	29	30	211	252	199	319	170	155	155	162	22.1
routes	9	2	4	161	84	77	434	21	20	33	33	74.0
breast	11	2	5	225	101	104	1954	24	22	49	51	78.2
echo	13	39	14	1208	6978	2773	1267	294	334	180	186	85.1
bridges	13	3	2	2417	5634	3343	2563	41	41	27	27	98.9
claims	13	1	1	14	4097	3086	4097	14	14	14	14	0.0
pdbx	13	11	5	2260	5409	3340	2799	97	99	59	72	97.4
adult	15	2	10	277	58	72	32722	31	27	109	112	53.4
hospital	15	12	4	10249	20420	15980	12364	142	143	66	75	99.4
lineitem	16	390	271	11986	53036	24492	13161	2660	3121	1894	2216	84.2
weather	18	523	230	105868	140602	115871	122295	3357	4077	2404	2803	97.7
ncvoter	19	113	42	232701	252281	249060	272163	1169	1403	509	539	99.8

with the border rather than with the levels of the lattice. *Takeaway*: the dualization step trades best-case efficiency for a worst case that is bounded by the border rather than by the lattice.

Real-world distributions. The opposite regime, a small border in a large lattice, is typical of real data. We use 12 real-world data sets with 9 to 19 columns, compute all minimal keys from their mined functional dependencies, and use them as the oracle: a question about X is answered ‘No’ iff X contains a minimal key. These keys are not meaningful, but they provide real-world distributions of minimal keys and maximal anti-keys, and an oracle that answers consistently and instantly, which makes it feasible to run all strategies to completion even when they need hundreds of thousands of questions. Tab. 2 shows that on 11 of the 12 data sets the best DA strategy asks fewer questions than the best LW strategy, and on *claims* the two tie at 14. On 10 data sets the reduction ranges from 53% to 99.8%. The exceptions are *claims*, whose border consists of one minimal key and one maximal anti-key, and *abalone*, whose border of 59 elements is almost as large as the levels of its 9-column lattice. The reduction is largest on the largest schema: on *ncvoter* the best LW strategy asks 232,701 questions, 44% of all column sets, while DA-BD asks 509. No data set has more than 753 border elements, while an LW strategy must traverse all levels between its starting point and the border. On *adult*, for instance, the two minimal keys contain 10 of the 15 columns each, and LW-BB asks 32,722 of the 32,768 column sets. No LW member dominates the others, as LW-TD, LW-BD, and LW-TB win on 8, 2, and 2 data sets, and their spread reaches a factor of 564 on *adult*, while the members of the DA family stay within a factor of 4.2 of each other on every data set. *Takeaway*: real-world schemata place a small border in a large lattice, which is exactly where the dualization-based family excels.

The smaller of the two bounds of Thm. 5.1 and Cor. 5.2 correctly suggests the better starting point of the DA family on every data set of Tab. 2: the top-down bound is smaller on the three data sets with only two minimal keys, where a top-down member wins, and on *abalone*, where the best members of both starting points tie; the bottom-up bound is smaller on the other data sets except *claims*,

where the bounds are equal and all members tie, and a bottom-up member wins on each of them. The bounds are also reasonably tight, as the best member asks between 37% and 94% of its bound. The direction matters less: depth-first and breadth-first differ by at most 22% in questions, while breadth-first saves up to a factor of 40 in the computation between questions, which matters once the border grows into the hundreds (App. B.1). Measured against the pace of human answers, this computation is negligible. On *weather*, the data set with the largest border, DA-BB computes for 3.9 seconds over 2,803 questions and DA-BD for 117 seconds over 2,404 questions, and even the slowest member, DA-TD, spends 30 minutes over 3,357 questions, about half a second per question, while the level-wise strategies spend up to 308 seconds on *lineitem*. Our human participants, in contrast, took between 27 and 95 seconds per question on average, depending on group and strategy (Tab. 4). This confirms in practice that the hard step of generating the next question is computed between two answers, at the pace of human deliberation. In answer to **RQ1**, the two families are complementary, which is why our suite contains both: LW strategies are preferable when the border is large compared to the lattice, while DA strategies are robust and preferable when a small border lies in the middle of a large lattice, as is typical of real-world schemata.

6.3 Comparison to Key Mining

For **RQ2**, we quantify the gap that i) the declared schema and ii) the state-of-the-art key mining tool *DataViadotto* [34] leave to the meaningful keys, on all 22 tables of the public Hockey database. Nine of the tables have no primary key, and no UNIQUE constraint is defined, despite several meaningful keys. The ground truth of meaningful keys on the Hockey tables is taken from a profiling study of the database by DataViadotto [?], in which every key was validated against the rules of ice hockey and against the documentation that explains the meaning of the tables and their columns. This is a reasonable way to fix a ground truth for meaningful keys: a column set is a rule exactly when the semantics of the domain forbids any two distinct records to agree on it, and the rules of the game together with the documented meaning of the columns determine this semantics independently of any data set and of any interviewee. Since this ground truth and the miner we compare against share their origin, any bias would favor the miner, so the gap we measure is, if anything, conservative. Our artifacts contain the ground truth together with the keys mined for all tables. We mined keys with up to four attributes (no meaningful key contains more), under possible and certain semantics for null markers [36], exactly and approximately with up to 1, 5, and 10% dirtiness (removing at most this percentage of records yields a valid key), under levels 0, 1, 5, and 9 of orthogonality [35], and with a completeness ratio of at least 0.75.

Tab. 3 shows the results. Orthogonality reduces false positives while retaining true ones [34]. Level L0 does not use it and thus represents mining that ignores meaningfulness [5]. At L0, precision and F1 are poor, especially for approximate mining, which trades a large loss in precision for a small gain in recall. Levels L1 to L9 raise precision at a small cost in recall. This comparison does not hold mining to a standard it was not designed for. With orthogonality, the miner is intended to return precisely the meaningful keys rather

Table 3: How Key Interviews Improve the Quality of Schema Specifications and Key Mining. Precision (P), recall (R) and F1 of the mined keys under possible and certain semantics, by percentage D of dirtiness and level L of orthogonality.

D	L	possible			certain		
		P	R	F1	P	R	F1
0	0	0.14	0.92	0.24	0.18	0.76	0.30
	1	0.29	0.79	0.43	0.26	0.66	0.37
	5	0.38	0.68	0.49	0.43	0.55	0.48
	9	0.62	0.63	0.62	0.56	0.50	0.53
1	0	0.03	0.95	0.05	0.03	0.79	0.06
	1	0.27	0.82	0.41	0.20	0.68	0.30
	5	0.38	0.71	0.49	0.42	0.58	0.49
	9	0.61	0.66	0.63	0.56	0.53	0.54
5	0	0.01	0.95	0.01	0.01	0.87	0.02
	1	0.24	0.82	0.38	0.16	0.74	0.26
	5	0.36	0.71	0.47	0.37	0.63	0.47
	9	0.60	0.66	0.62	0.56	0.58	0.57
10	0	0.004	0.95	0.01	0.005	0.92	0.01
	1	0.21	0.82	0.33	0.13	0.79	0.23
	5	0.34	0.71	0.46	0.30	0.66	0.42
	9	0.60	0.66	0.62	0.56	0.61	0.58
primary keys on schema: P 0.92, R 0.32, F1 0.47							

than all keys that happen to hold on the data, so levels L1 to L9 represent the best current attempt to infer meaningfulness from data alone. Still, mining performs poorly in every setting: averaged over the settings, F1 is 0.241 (possible) and 0.297 (certain) for exact mining at L0, only 0.025 and 0.030 for approximate mining at L0, and 0.513 and 0.459 for exact mining with orthogonality. The declared primary keys reach a precision of 0.92 but a recall of only 0.32. Hence, mined keys miss between 48.7 and 97.5% of the F1 of the meaningful keys, which is the gap our methodology aims to close by putting domain experts in charge of the decision. That even the miner designed for meaningfulness reaches an F1 of at most 0.63 shows the limits of data as the source of truth, and the risk of declaring mined keys as rules. Since this ground truth is independent of our interviews, Secs. 6.4 and 6.5 can measure how much of the gap human and artificial interviewees close. *Takeaway:* data can suggest candidates, but deciding which keys are rules requires domain knowledge.

6.4 Human Interviewees of Different Expertise

For RQ3, we interviewed ten human participants on the Hockey tables for which the ground truth is available. The study is designed to provide first insight into how humans of different expertise use our interviews, rather than a definitive evaluation. We found our participants to be generally generous with their time.

Participants. Before the study, and independently of any outcome, we assigned the ten participants to three groups according to their job expertise and their experience with data and with the domain of ice hockey. *Domain experts* (E1–E3) work with data professionally, as analysts or engineers with at least five years of experience in data modeling or analytics, and have followed professional ice hockey and its statistics in detail for many years. *Data practitioners* (P1–P4) are software and data professionals or data-management researchers with two to five years of experience, who are fluent with the concept of a key but follow ice hockey only casually. *Non-specialists* (N1–N3) work with data only occasionally, for example with spreadsheets in business or administrative roles

or as students early in a computing degree, and know little about ice hockey.

Protocol. Participants were recruited by invitation and took part voluntarily and without compensation, after giving informed consent. They were told in advance that the tool records each answer and the time taken to give it, that they could stop at any point or decline any question without giving a reason, and that their records would be reported only in aggregated and pseudonymized form. All participants used the online interview tool of our artifact. For each of nine Hockey tables, they were shown the schema and acquired its minimal keys by answering the questions of each of the eight strategies, presented in randomized order, without access to the ground truth, to the LLM results, or to our hypotheses. Interviews were run on the prime attributes of each table as defined by the ground truth (Sec. 6.5) and repeated several times to assess the stability of the answers. On the four tables with fewer than 11 columns, every strategy was also run on the full schema with a budget of half an hour of participant time.

Table 4: Ten human participants in three groups of job expertise and experience, on the nine Hockey tables for which the ground truth of meaningful keys is available. Four tables of schema size below 11 are used for the full-schema experiments. The methods TD to BB are the four traversals of the family named above them, so that BB in the left half is LW-BB and BB in the right half is DA-BB; Agg and M2 to M4 aggregate the four traversals of that family.

	Method	level-wise (LW)						dualize-and-advance (DA)							
		Exact			prime filt.		full schema	Exact			prime filt.		full schema		
		F1	R	P	#Q	min	done	#Q	F1	R	P	#Q	min	done	#Q
domain experts (3)	TD	0.91	0.90	0.92	4.3	3.2	12/12	11	0.91	0.90	0.92	4.3	3.2	12/12	10
	TB	0.91	0.90	0.92	4.2	3.1	12/12	25	0.91	0.90	0.92	4.3	3.2	12/12	10
	BD	0.85	0.84	0.86	6.0	3.5	12/12	31	0.87	0.86	0.88	7.5	3.9	12/12	24
	BB	0.80	0.80	0.80	9.8	4.4	1/12	104	0.87	0.86	0.88	7.7	3.9	12/12	25
	Agg	0.80	0.80	0.80	—	—	—	—	0.87	0.86	0.88	—	—	—	—
	M2	0.85	0.84	0.86	—	—	—	—	0.87	0.86	0.88	—	—	—	—
	M3	0.85	0.84	0.86	—	—	—	—	0.87	0.86	0.88	—	—	—	—
	M4	0.80	0.80	0.81	—	—	—	—	0.87	0.86	0.88	—	—	—	—
data practitioners (4)	TD	0.59	0.63	0.56	5.2	6.1	16/16	13	0.58	0.62	0.56	5.1	6.1	16/16	11
	TB	0.58	0.63	0.56	5.2	6.2	16/16	24	0.58	0.63	0.56	5.1	6.1	16/16	11
	BD	0.57	0.61	0.54	7.3	7.0	16/16	26	0.53	0.57	0.51	8.9	7.6	16/16	20
	BB	0.50	0.54	0.48	13.5	9.6	2/16	55	0.53	0.56	0.51	9.4	7.8	16/16	21
	Agg	0.50	0.54	0.48	—	—	—	—	0.53	0.56	0.51	—	—	—	—
	M2	0.57	0.61	0.55	—	—	—	—	0.53	0.57	0.51	—	—	—	—
	M3	0.57	0.61	0.55	—	—	—	—	0.53	0.57	0.52	—	—	—	—
	M4	0.51	0.53	0.50	—	—	—	—	0.53	0.56	0.52	—	—	—	—
non-specialists (3)	TD	0.36	0.37	0.36	5.3	8.1	10/12	14	0.36	0.37	0.37	5.2	8.1	12/12	10
	TB	0.36	0.37	0.36	5.1	8.0	8/12	30	0.36	0.37	0.37	5.2	8.2	12/12	11
	BD	0.33	0.34	0.34	6.7	8.9	8/12	32	0.32	0.33	0.33	7.9	9.6	12/12	23
	BB	0.31	0.31	0.31	11.0	11.3	0/12	46	0.32	0.33	0.33	8.2	9.7	10/12	23
	Agg	0.31	0.31	0.31	—	—	—	—	0.32	0.33	0.33	—	—	—	—
	M2	0.33	0.34	0.34	—	—	—	—	0.32	0.33	0.33	—	—	—	—
	M3	0.34	0.34	0.35	—	—	—	—	0.33	0.33	0.33	—	—	—	—
	M4	0.31	0.31	0.32	—	—	—	—	0.33	0.33	0.33	—	—	—	—

What the numbers measure. Since the human interviews ran on the prime attributes defined by the ground truth, Tab. 4 measures how well humans acquire the meaningful minimal keys once the relevant attributes are known. It does not measure the benefit of the interview over an alternative procedure, which our study was not designed to isolate. Within this setting, each question asks for exactly the judgment that minimality requires. Adding columns only increases the confidence that a column set identifies a record, whereas establishing that a column is removable requires

considering a proper subset, which is what a traversal puts to the interviewee. *Takeaway:* the results show how much of the gap humans close with our interviews when the relevant attributes are known, not a comparison with another procedure.

Expertise determines how much of the gap is closed. The groups reach an F1 of 0.91, 0.59 and 0.36 under their best strategy, and the worst strategy of a domain expert, at 0.80, still exceeds the best strategy of a data practitioner by 0.21. The gaps between the groups, 0.32 and 0.23, exceed every gap between strategies within a group, which ranges from 0.05 to 0.11. Our suite guarantees that every candidate is considered, so the acquired keys are those that the answers determine; Tab. 4 measures how far these answers are from the ground truth. Even domain experts fall short of perfect precision and recall, which confirms that it is an aim the interview works towards, and that the expertise brought to it decides how far it gets. The direction of errors matches the profiles of the groups. Domain experts omit rather than invent keys, so their precision exceeds their recall under every strategy except LW-BB, where the two are equal. Data practitioners claim uniqueness for column sets that do not have it, so their recall exceeds their precision by 0.05 to 0.07, which is consistent with recognizing identifying column sets in typical data without knowing which records may legitimately coincide in the domain. Non-specialists miss about as many keys as they invent. *Takeaway:* the interview cannot create expertise, but it helps every interviewee make full use of the expertise they have.

Fewer questions, fewer inconsistencies. A consistent interviewee would return the same keys under all eight strategies. Humans are not perfectly consistent, and every additional question is an opportunity to contradict an earlier judgment. Accordingly, F1 decreases as the number of questions grows in every family and group: for domain experts it falls from 0.91 at 4.2 questions (LW-TB) to 0.85 at 6.0 (LW-BD) and 0.80 at 9.8 (LW-BB). The DA family is less affected, as its members fall only from 0.91 to 0.87. This suggests a top-down strategy as the default for humans, since in every group the top-down strategies match the best F1 to within 0.01 while asking the fewest questions. Aggregating the four traversals of a single participant does not help: *Agg* reproduces the bottom-up strategies, and the best frequency-based aggregate, *M3*, reaches 0.85 against 0.91 for LW-TB at four times the cost. *Takeaway:* for human interviewees, fewer questions mean not only less effort but also higher quality.

Aggregating answers across the team. Tab. 5 accepts a key once at least k of the ten participants have acquired it. Both families peak at $k = 4$, at an F1 of 0.85 (LW) and 0.90 (DA), where the accepted keys equal the ground truth on 8.0 of the nine tables. A team in which only three of ten members are domain experts thus reaches the quality of a single domain expert, 0.90 against 0.91, without knowing who the experts are. Both sides of the peak fail: below it, recall exceeds precision and the team accepts keys it should not; above it, precision exceeds recall until nothing survives at $k = 10$. The permissive side is worse, as the union ($k = 1$) reaches only 0.13: a participant who misses a distinction between records contributes a proper subset of a correct key, and minimization then eliminates the correct key. The DA family is ahead at every threshold except $k = 3$ (tie) and $k = 10$, since its traversals dissent less. In answer to **RQ3**, given the relevant attributes, domain experts close most of the gap, expertise rather than strategy determines

Table 5: Resolving different answers within the team of ten participants. A key is accepted when at least k participants acquire it. The experiment uses the prime-filtered setting, so that the threshold captures differences in the participants’ domain knowledge. Results are averaged over the nine Hockey tables and over the repeated interviews, separately for the four traversals of each family; *solved* is the mean number of those nine tables on which the accepted key set equals the ground truth.

	k	1	2	3	4	5	6	7	8	9	10
LW	F1	0.09	0.43	0.66	0.85	0.79	0.50	0.21	0.09	0.07	0.00
	Recall	0.09	0.46	0.67	0.84	0.78	0.49	0.20	0.09	0.07	0.00
	Precision	0.09	0.41	0.65	0.87	0.82	0.53	0.22	0.09	0.07	0.00
	solved	0.8	3.0	5.4	7.4	6.6	4.0	1.6	0.8	0.6	0.0
DA	F1	0.13	0.50	0.66	0.90	0.87	0.61	0.28	0.13	0.08	0.00
	Recall	0.13	0.52	0.68	0.90	0.86	0.59	0.27	0.12	0.08	0.00
	Precision	0.12	0.48	0.65	0.91	0.89	0.64	0.31	0.16	0.09	0.00
	solved	1.0	3.6	5.2	8.0	7.4	4.8	2.0	0.8	0.6	0.0

how much of it is closed, and a moderate majority of a mixed team recovers the quality of a domain expert.

Threats to validity. As a study for first insight, ten participants, a single database, and group assignments based on job profiles limit what the per-group averages support. Every participant answered the same tables under all eight strategies; since the strategies were presented in randomized order, a practice effect is spread across the strategies rather than favoring any of them, although it may raise the overall quality. Our participants were generous with their time, and volunteers may persist longer and try harder than an arbitrary interviewee; we withheld the ground truth, the hypotheses, and all feedback until every task was complete, but the domain-expert quality should be read as an upper bound. The prime attributes of the prime-filtered human interviews were defined by the ground truth, so these interviews show what humans achieve once the relevant attributes are known; with predicted prime attributes, quality may be lower, although the LLM predictions of Sec. 6.5 never missed a prime attribute. Finally, the sports domain is publicly documented, which may overstate what an expert of a proprietary schema achieves and understate what a non-specialist does.

6.5 Focusing on Prime Attributes

For **RQ4**, we study the heuristic of restricting an interview to the *prime attributes* of a table, that is, the attributes that occur in some meaningful minimal key. This reduces the search space from $2^{|T|}$ to $2^{|P|}$ column sets for the set P of attributes selected as prime, and in practice meaningful keys are few and small. We let two open-source LLMs, DeepSeek-R1-Distill-Llama-70B (DS) and Qwen3-30B-A3B-Thinking-2507 (Qwen), predict the prime attributes of a table, and then act as interviewees on the reduced schema. The prediction is the only step at which completeness is traded for scalability: our suite still considers every candidate over P , but a meaningful key that contains an attribute outside P cannot be acquired. App. B.2 reports the LLM interviews in full.

LLM interviewees. Tab. 6 compares full-schema with prime-filtered interviews. On full schemata, Qwen acquires exactly the meaningful keys under all eight strategies on two of the four tables,

Table 6: Effect of prime filtering on LLM interviewees for the Hockey tables. F1 is the exact F1 averaged over the eight strategies, #1 the number of strategies with an F1 of 1, and #Q the average number of questions per strategy. $|P|$ is the number of predicted prime attributes and $F1_P$ the F1 of this prediction. Full-schema interviews were run on the tables with $|T| \leq 8$.

table	$ T $	LLM	full schema			$ P $	prime filtering				
			F1	#1	#Q		$F1_P$	F1	#1	#Q	
Scoring Shootout	7	DS	0.21	0	35.0	3	1.00	1.00	8	5.5	
		Qwen	1.00	8	25.5	3	1.00	1.00	8	5.5	
Goalies Shootout	8	DS	0.40	2	60.6	3	1.00	1.00	8	5.5	
		Qwen	0.67	2	62.1	3	1.00	1.00	8	5.5	
Team Vs Team	8	DS	0.31	0	57.6	4	0.86	1.00	8	8.2	
		Qwen	1.00	8	54.5	4	0.86	1.00	8	7.8	
Combined Shutouts	8	DS	0.20	0	53.8	5	1.00	0.25	2	13.2	
		Qwen	0.51	0	54.5	5	1.00	0.59	1	13.1	
Teams Half	11	DS	–	–	–	3	1.00	1.00	8	5.5	
		Qwen	–	–	–	3	1.00	1.00	8	5.5	
Teams Post	17	DS	–	–	–	3	0.80	0.83	4	5.4	
		Qwen	–	–	–	2	1.00	1.00	8	3.5	
Goalies	23	DS	–	–	–	4	0.86	0.67	0	9.4	
		Qwen	–	–	–	3	1.00	1.00	8	5.5	
Team Splits	43	DS	–	–	–	3	0.80	1.00	8	5.0	
		Qwen	–	–	–	2	1.00	1.00	8	3.5	

while DS does so on none of them and only under two strategies on one table, so the quality of LLM interviewees depends strongly on the model (App. B.2). The predictions of prime attributes are good: they always have perfect recall, and their imprecision only adds superfluous attributes. On the four tables with full-schema interviews, prime filtering reduces the number of questions over all strategies from 3,229 to 515, by 84%, and the interview time of the LLMs from 6,177 to 681 minutes. It also raises quality, from an average F1 of 0.28 to 0.81 for DS and from 0.79 to 0.90 for Qwen. For example, DS fails on Scoring Shootout and Team Vs Team on the full schema but achieves perfect scores under all eight strategies after filtering, since irrelevant attributes no longer distract it. On larger tables, schemata of up to 43 columns are reduced to two to four attributes, and on six of the eight pairs of table and model every strategy acquires exactly the meaningful keys. Prime filtering does not repair every error, however. On Combined Shutouts, both models predict the prime attributes perfectly but still answer some questions over the five remaining attributes wrongly, and on Goalies DS accepts a wrong key over its four predicted attributes, one of which is superfluous. *Takeaway:* prime filtering removes most of the questions an LLM can get wrong, but not the errors it makes on the questions that remain.

Scaling to large schemata. On the 12 real-world data sets of Tab. 2, which lack a ground truth, a prime-filtered LLM interview asks between 2.0 and 12.4 questions per strategy on average over prime-attribute budgets $|P| = 1, \dots, 5$ (App. B.2), against up to 272,163 questions on the full schema. Summed over all data sets and both models, the DA family asks 524.2 questions against 637.2 of the LW family. *Takeaway:* with prime filtering, the size of the schema no longer determines the effort of an interview.

Human interviewees. Prime filtering has the same effect on humans. All prime-filtered human interviews in Tab. 4 finish, with 4.2 to 13.5 questions and 3.1 to 11.3 minutes of participant time per traversal. On the full schema, in contrast, only 271 of 320 traversals

finish within half an hour, 113 of 160 for the LW family and 158 of 160 for the DA family, and LW-BB completes only 3 of 40. Prime filtering reduces the number of questions of every strategy in every group, by a factor between 1.9 and 10.6, where the largest factors are lower bounds since unfinished traversals understate their cost. The effort of an interview thus becomes small and predictable, while the same eight strategies range from 10 to more than 104 questions on the full schema. The Hockey database is public, so its schema and documentation may be part of the LLMs' training data, which could inflate their quality; this concern does not apply to the effort measured on the 12 data sets without ground truth, and it underlines that LLMs should aid rather than replace human experts. In answer to **RQ4**, focusing on prime attributes is what scales our interviews to large schemata, for artificial and human interviewees alike, and for LLMs it also improves quality, provided that no prime attribute is missed.

7 Conclusion and Future Work

Identifying the meaningful minimal keys with perfect precision and recall is a prerequisite for correct data modeling, data integrity, and efficient data management. It is also hard: we showed that enumerating the interview questions is as hard as enumerating the edges of a transversal hypergraph. We proposed a human-centered methodology that aims to close the gap towards perfect precision and recall, and two families of algorithms that generate Boolean questions for experts while guaranteeing that no candidate is overlooked, so that the acquired keys are exactly those that the experts' answers determine. Our experiments show that the two families complement each other across distributions of minimal keys, with the dualization-based family asking a number of questions proportional to the border of minimal keys and maximal anti-keys and saving up to 99.8% of the questions on real-world data. They show that key mining leaves a gap of 48.7 to 97.5% in F1, which calls for domain experts as decision makers. Our interview study shows that, given the relevant attributes, the three groups of expertise reach an F1 of up to 0.91, 0.59 and 0.36, that expertise rather than strategy determines how much of the gap is closed, and that a team of ten of which three are domain experts recovers the quality of a domain expert once four of them accept a key. Finally, focusing interviews on prime attributes predicted by LLMs makes them feasible on large schemata and, for LLM interviewees, also improves quality.

Perfect precision and recall thus remains the aim rather than a guaranteed outcome, and future work will narrow the remaining gap along the dimensions of research in Fig. 6 (App. C): more efficient traversal strategies; an extension to foreign keys, whose search space can be reduced to references to the meaningful keys acquired; support for teams whose members disagree or cannot answer a question; and domain-specific training of LLMs as interviewees.

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A Proofs

PROOF OF THM. 4.1. We show that the known $W[1]$ -complete problem CLIQUE [16, Thm. 21.2.4] fpt-reduces to ANTI-KEY. A graph $G = (V, E)$ and number $k \in \mathbb{N}$ are in CLIQUE, iff G contains a clique of size k . Given an instance (G, k) , construct an instance $f(G, k)$ of ANTI-KEY, by setting the table schema T to be the vertices V of the graph, the set of keys C to be the edges *not* in E , and keeping the parameter k as is. We now have a one-to-one-correspondence between anti-keys and cliques. If G contains a clique of size k then $f(G, k)$ has an anti-key for C over T of the same size and *vice versa*. The function f is computable even in polynomial-time yielding $W[1]$ -hardness as desired. $\square$

PROOF OF THM. 4.2. A column set $A \subseteq T$ is an anti-key for C iff A contains no $K \in C$, iff $T \setminus A$ intersects every $K \in C$, that is, iff $T \setminus A$ is a transversal of $\mathcal{H}_C = (T, C)$. As complementation reverses inclusion, A is a maximal anti-key iff $T \setminus A$ is a minimal transversal, so complementation is a polynomial-time bijection between the solutions of ANTI-KEY-ENUM on C and of HYPERGRAPH-TRANSVERSAL-ENUM on $\mathcal{H}_C$, and conversely every hypergraph $\mathcal{H} = (V, E)$ is of the form $\mathcal{H}_C$ for $T = V$ and $C = E$. For ARMSTRONG-ENUM, there is a one-to-one correspondence between the edges of the transversal hypergraph and the records of the Armstrong table beyond a fixed base record, as described in Sec. 4 [23, p. 3]. $\square$

PROOF OF THM. 5.1. Every question concerns a column set that is not settled, that is, a column set whose status is not implied by previous answers or input keys. If a candidate of the dualization step is answered ‘No’, it is a minimal key by the argument of Sec. 5.4 and, once in R , is settled and never questioned again, so at most $|\mathcal{K}|$ questions are answered ‘No’ in a dualization step. If a candidate is answered ‘Yes’, it is a counterexample, and the advance step of Alg. 3 asks at most $|T \setminus X| \leq |T|$ further questions for it. The advance step turns the counterexample into a maximal anti-key: no further column can be added without contradicting an answer or an input key. Since the counterexample is not $\leq$ -below any member of B , each advance step adds a new maximal anti-key to B . Hence, at most $|\mathcal{K}^{-1}|$ candidates are answered ‘Yes’, each followed by at most $|T|$ questions, which also bounds the number of rounds, and the claim follows. $\square$

PROOF OF COR. 5.2. By duality, the argument of Thm. 5.1 applies with the roles of minimal keys and maximal anti-keys exchanged: every candidate confirmed as an anti-key is one of the $|\mathcal{K}^{-1}|$ maximal anti-keys and never questioned again, and every counterexample is shrunk to a new minimal key with at most $|T|$ further questions. $\square$

B Additional Experiments

B.1 Distributions of Minimal Keys

We complement Sec. 6.2 with further observations on the synthetic and real-world distributions of minimal keys.

Depth-first level-wise strategies. Under probabilistic answering (Fig. 5), LW-TD peaks between $p = 0.3$ and $p = 0.4$ and LW-BD between $p = 0.6$ and $p = 0.8$. Consistently across both schema sizes, LW-TD is ahead at $p = 0$, LW-BD from $p = 0.2$ to $p = 0.5$, LW-TD again from $p = 0.6$ to $p = 0.8$, and LW-BD at $p = 1$. The

two crossings in between move outwards as the schema grows: for $|T| = 11$ they lie at $p = 0.2$ and $p = 0.8$, while for $|T| = 15$ LW-BD is already ahead at $p = 0.1$ and LW-TD is still ahead at $p = 0.9$.

The direction. The two directions differ by at most 22% in the number of questions, on *pdxb*, where DA-BD asks 59 questions and DA-BB asks 72, and by up to a factor of 40 in the computation that the tool performs between the questions. For the bottom-up members, depth-first is never worse, asking fewer questions on 9 data sets and the same number on the other 3, while for the top-down members the two directions split, with depth-first ahead on 6 data sets, breadth-first on 4, and a tie on 2. The computational cost ranks the two directions in the opposite order. A depth-first member advances the first counterexample of a round immediately and therefore pays one dualization per border element, whereas a breadth-first member amortizes one dualization over all counterexamples of a round. On *weather*, the data set with the largest border, DA-BD spends 117 seconds of computation for 2,404 questions while DA-BB spends 3.9 seconds for 2,803 questions, and DA-TD spends 30 minutes for 3,357 questions while DA-TB spends 44 seconds for 4,077 questions. Since this computation happens between the questions and consumes no effort of the experts, depth-first remains the default. Once the border grows into the hundreds, however, breadth-first becomes the direction of choice, since it amortizes the dualization time and lets a team of experts answer the questions of a round in parallel. For comparison, the level-wise strategies spend up to 58 seconds on *hospital*, where every dualization-based strategy finishes within 11 milliseconds, and up to 308 seconds on *lineitem*.

Which questions are asked. The two families also differ in the answers that the experts give, which our runs record. A level-wise strategy spends nearly all of its questions on one side of the border, as LW-TB and LW-BD answer ‘No’ to almost every question, on *hospital* to 20,416 of 20,420 and to 15,962 of 15,980, while LW-TD and LW-BB answer ‘Yes’ to all but 25 of 10,249 and all but 12 of 12,364 on the same data set. The dualization-based strategies keep both answers informative, since the best member of a data set receives ‘No’ to between 18% and 65% of its questions, with the sole exception of *claims*, whose border consists of a single minimal key and a single maximal anti-key.

B.2 LLMs as Domain Experts

LLMs are experts of natural language and of many domains, so it is natural to ask how well they can replace domain experts in our interviews, in particular when human experts are not available or to complement their expertise. We use DeepSeek-R1-Distill-Llama-70B (DS) and Qwen3-30B-A3B-Thinking-2507 (Qwen) in two settings: (1) *direct LLM interviews*, where the LLM answers the questions generated over the full schema, and (2) *prime filtering*, where the LLM first predicts the prime attributes and the interview is restricted to them (Sec. 6.5). The aggregates *Agg.* and *Mk* range over the four strategies of a single family, which the prefixes LW and DA identify.

We first evaluate the quality of LLM-based interviews on the Hockey database, where the ground truth is available as its set of meaningful keys. Tab. 7 reports the results of direct LLM interviews on tables with relatively small schema sizes. Specifically,

it reports both exact and relaxed F1, Recall, and Precision scores under different traversal strategies. For relaxed scores, we regard an LLM-returned key as correct for precision if it is a superkey of some ground-truth key. Similarly, we regard a ground truth key as recovered for recall if it is a superkey of some LLM-returned key. Overall, Qwen is substantially more reliable than DS. For example, Qwen achieves perfect exact and relaxed scores on several tables and strategies, such as Scoring Shootout and Team Vs Team, while DS is unstable and fails completely on some tables, such as Scoring Shootout. This indicates that LLMs can provide useful semantic judgment for key acquisition, but their reliability is strongly model-dependent. The results also show that traversal strategies matter. Breadth-first and depth-first variants often require different numbers of questions, and return different results. This may be regarded as digital equivalent to a difference in opinion between human experts. Here, we aggregate results by i) selecting all keys that are minimal across all traversal strategies (Agg), and ii) keeping only keys returned by at least k traversal strategies (Mk for $k = 1, \dots, 4$). Aggregation does not always improve accuracy because it may accumulate false positives from different traversals. Frequency-based minimization Mk can sometimes correct false positives, but its effectiveness depends on the quality and consistency of the keys returned.

Tab. 8 evaluates prime filtering in terms of the ground truth. This strategy reduces the search space from $2^{|T|}$ to $2^{|P|}$, where P is the set of prime attributes predicted by LLMs. Intuitively, $|P|$ is much smaller than $|T|$, so if LLMs can predict the prime attributes well, then this optimizes the efficiency of key acquisition while maintaining a high quality.

The light-blue rows in Tab. 8 use the same Hockey tables as Tab. 7, allowing us to compare the two LLM-based strategies: direct interviews over the original schema versus prime filtering. The comparison shows that prime filtering can substantially improve both efficiency and robustness. For example, on Scoring Shootout, DS fails in the direct interview setting in Tab. 7, but achieves perfect scores after the schema is reduced to the predicted prime attributes in Tab. 8. Similar improvements are observed on Team Vs Team, where DS changes from mixed or failed results in the direct setting to perfect results after filtering. In addition, the number of questions is greatly reduced across these small-schema tables. For instance, on Goalies Shootout, direct interviewing requires dozens to hundreds of questions depending on the model and strategy, while the reduced-schema setting requires only a small number of questions for the individual traversal strategies. This suggests that even when the original schema is not too large, prime filtering can remove irrelevant attributes and make the downstream interview process easier for LLMs.

The light-orange rows in Tab. 8 further demonstrate the benefit of prime filtering on larger Hockey tables. In our experiments, large schemata are often reduced to few prime attributes. For instance, Team Splits has $|T| = 43$, but DS and Qwen reduce it to three and two prime attributes, respectively, while still achieving perfect results. Similar reductions hold for Teams Post and Goalies, where the filtering step substantially lowers the number of interview questions while preserving high exact and relaxed scores.

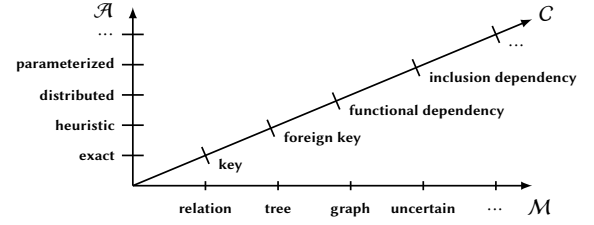


Figure 6: Dimensions of Research in Constraint Acquisition

We will now evaluate the scalability on various real-world data sets where the original interview process may require a large number of questions. Tab. 9 reports the results of prime filtering on these data sets. In the absence of any ground truth, we cannot report on precision or recall. Instead, Tab. 9 focuses on the number of questions generated and the runtime required by LLM-based interviews. The results show that LLMs complete the interview process with a feasible number of questions and reasonable runtime, even for data sets where the original search space is large. This pattern is due to the number of likely prime attributes being small, which is consistent with observations in practice.

Indeed, every entry of Tab. 9 is the average over all prime-attribute budgets $|P| = 1, \dots, 5$. So even for larger numbers of prime attributes than predicted, the interview process is very feasible in the number of questions and time spent. Summed over the 24 rows of Tab. 9, the dualization-based family asks 524.2 questions in 544.5 minutes against 637.2 questions in 693.4 minutes of the level-wise family, and it asks fewer questions on 21 of the 24 rows; the exceptions are abalone and weather under DS and echo under Qwen, where it asks between 2.0 and 2.6 questions more. As an indicator for comparison, Tab. 2 reports the number of questions that the same strategies generate on the full schema of these data sets. Since the process would take too long for LLMs to complete in this case, we answered “No” iff the key in question is a super-key of some minimal key mined from the data set. Tabs. 7 and 8 complement this scalability evidence by showing that, when the ground truth is available, LLM-based interviews can also achieve strong prediction performance, especially when prime filtering is used before the interview. In particular, prime filtering is effective in making the interview process efficient while maintaining high precision and recall. These results suggest that LLMs are most useful when they are integrated as a scalable aid to human-centered key acquisition: they can reduce the practical burden of the interview process, while human expertise remains important for validating final decisions in high-stakes settings.

C Dimensions of Research

The methodology gives scope to a rich landscape of research directions. One may select any combination of dimensions from algorithmic approaches $\mathcal{A}$, the class $\mathcal{C}$ of constraints, and the model $\mathcal{M}$ of data. Each combination defines a specific research setting in this area, as illustrated in Fig. 6.

Algorithmic approaches may consider traversal strategies such as bottom-up, top-down, depth-first, breadth-first, or mixed ones; but also exact, heuristic, approximate, randomized, distributed and parameterized algorithms. Classes of constraints may include keys,

Table 7: LLM-based interview results for tables in the Hockey database on *full schemata*. Abbreviations: Agg. = aggregation of all interviewed minimal keys from all strategies; Mk = minimization of interviewed keys with frequency threshold k . An aggregate ranges over the four strategies of one family, which the prefixes LW and DA identify. Rows are shaded according to schema size: light blue for $|T| \leq 10$ in Tabs. 7 & 8 and light orange for $|T| > 10$ in Tab. 8.

Table	Model	T	Strat.	Exact F/R/P	Relaxed F/R/P	#Q	Time min	Table	Model	T	Strat.	Exact F/R/P	Relaxed F/R/P	#Q	Time min		
Scoring Shootout	DS	7	LW-TD	0.00/0.00/0.00	0.00/0.00/0.33	30	53.18	Goalies Shootout	DS	8	LW-TD	0.50/1.00/0.33	0.50/1.00/0.33	99	155.61		
			LW-TB	0.00/0.00/0.00	0.00/0.00/0.75	18	31.60				LW-TB	1.00/1.00/1.00	1.00/1.00/1.00	35	49.47		
			LW-BD	0.00/0.00/0.00	0.00/0.00/0.75	24	39.02				LW-BD	1.00/1.00/1.00	1.00/1.00/1.00	27	39.92		
			LW-BB	0.00/0.00/0.00	0.00/1.00/0.00	93	165.79				LW-BB	0.50/1.00/0.33	0.50/1.00/0.33	195	279.94		
			LW Agg.	0.00/0.00/0.00	0.50/1.00/0.33	165	289.59				LW Agg.	0.50/1.00/0.33	0.50/1.00/0.33	356	524.94		
			LW M4	0.00/0.00/0.00	0.00/0.00/0.00	-	-				LW M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-		
			LW M3	0.00/0.00/0.00	0.00/0.00/0.67	-	-				LW M2	0.50/1.00/0.33	0.50/1.00/0.33	-	-		
			LW M2	0.00/0.00/0.00	0.00/0.00/0.50	-	-				DA-TD	0.00/0.00/0.00	0.00/0.00/1.00	10	13.19		
			LW M1	0.00/0.00/0.00	0.00/1.00/0.00	-	-				DA-TB	0.00/0.00/0.00	0.00/0.00/1.00	10	13.17		
			DA-TD	0.50/1.00/0.33	0.50/1.00/0.33	19	24.53				DA-BD	0.22/1.00/0.13	0.22/1.00/0.13	55	82.51		
			DA-TB	0.50/1.00/0.33	0.50/1.00/0.33	19	24.52				DA-BB	0.00/0.00/0.00	0.00/1.00/0.00	54	79.58		
			DA-BD	0.33/1.00/0.20	0.33/1.00/0.20	38	55.92				DA Agg.	0.18/1.00/0.10	0.33/1.00/0.20	129	188.45		
			DA-BB	0.33/1.00/0.20	0.33/1.00/0.20	39	56.92				DA M2	0.00/0.00/0.00	0.00/0.00/0.00	-	-		
			DA Agg.	0.33/1.00/0.20	0.33/1.00/0.20	115	161.88				DA M1	0.00/0.00/0.00	0.00/1.00/0.00	-	-		
			DA M4	0.50/1.00/0.33	0.50/1.00/0.33	-	-				LW-TD	0.67/1.00/0.50	0.67/1.00/0.50	99	234.55		
			DA M2	0.33/1.00/0.20	0.33/1.00/0.20	-	-				LW-TB	1.00/1.00/1.00	1.00/1.00/1.00	35	161.11		
	Qwen	7	LW-TD	1.00/1.00/1.00	1.00/1.00/1.00	8	14.72		Qwen	8	LW-BD	1.00/1.00/1.00	1.00/1.00/1.00	42	174.86		
			LW-TB	1.00/1.00/1.00	1.00/1.00/1.00	19	44.45				LW-BB	0.67/1.00/0.50	0.67/1.00/0.50	222	493.82		
			LW-BD	1.00/1.00/1.00	1.00/1.00/1.00	10	14.49				LW Agg.	0.67/1.00/0.50	0.67/1.00/0.50	398	1064.34		
			LW-BB	1.00/1.00/1.00	1.00/1.00/1.00	113	199.97				LW M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-		
			LW Agg.	1.00/1.00/1.00	1.00/1.00/1.00	150	273.63				LW M2	0.67/1.00/0.50	0.67/1.00/0.50	-	-		
			LW M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-				DA-TD	0.67/1.00/0.50	0.67/1.00/0.50	18	53.57		
			DA-TD	1.00/1.00/1.00	1.00/1.00/1.00	8	15.35				DA-TB	0.00/0.00/0.00	0.00/0.00/0.50	19	57.91		
			DA-TB	1.00/1.00/1.00	1.00/1.00/1.00	8	13.69				DA-BD	0.67/1.00/0.50	0.67/1.00/0.50	31	87.31		
DA-BD			1.00/1.00/1.00	1.00/1.00/1.00	19	33.36	DA-BB	0.67/1.00/0.50			0.67/1.00/0.50	31	87.10				
DA-BB			1.00/1.00/1.00	1.00/1.00/1.00	19	33.68	DA Agg.	0.40/1.00/0.25			0.67/1.00/0.50	99	285.89				
DA Agg.			1.00/1.00/1.00	1.00/1.00/1.00	54	96.09	DA M3	1.00/1.00/1.00			1.00/1.00/1.00	-	-				
DA M4			1.00/1.00/1.00	1.00/1.00/1.00	-	-	DA M2	0.67/1.00/0.50			0.67/1.00/0.50	-	-				
DS			8	LW-TD	0.50/1.00/0.33	0.50/1.00/0.33	65	101.80			DS	8	LW-TD	0.00/0.00/0.00	0.00/1.00/0.00	58	89.83
				LW-TB	0.00/0.00/0.00	0.00/0.00/1.00	19	29.93					LW-TB	0.40/1.00/0.25	0.40/1.00/0.25	54	82.46
	LW-BD	0.00/0.00/0.00		0.00/0.00/1.00	20	30.45	LW-BD	0.40/1.00/0.25	0.40/1.00/0.25	56			92.03				
	LW-BB	0.50/1.00/0.33		0.50/1.00/0.33	217	308.26	LW-BB	0.00/0.00/0.00	0.00/1.00/0.00	80			112.41				
	LW Agg.	0.20/1.00/0.11		0.88/1.00/0.78	-	-	LW Agg.	0.12/1.00/0.06	0.12/1.00/0.06	248			376.74				
	LW M2	0.50/1.00/0.33		0.50/1.00/0.33	-	-	LW M2	0.00/0.00/0.00	0.00/1.00/0.00	-			-				
	DA-TD	0.50/1.00/0.33		0.50/1.00/0.33	26	32.02	LW M1	0.00/0.00/0.00	0.00/1.00/0.00	-			-				
	DA-TB	0.00/0.00/0.00		0.00/0.00/0.50	26	31.17	DA-TD	0.67/1.00/0.50	0.67/1.00/0.50	18			26.53				
	DA-BD	0.50/1.00/0.33		0.50/1.00/0.33	43	51.59	DA-TB	0.00/0.00/0.00	0.00/0.00/0.50	18			28.66				
	DA-BB	0.50/1.00/0.33		0.50/1.00/0.33	45	54.02	DA-BD	0.13/0.50/0.07	0.13/1.00/0.07	73			108.86				
	DA Agg.	0.40/1.00/0.25		0.67/1.00/0.50	140	168.81	DA-BB	0.00/0.00/0.00	0.00/1.00/0.00	73			103.46				
	DA M4	0.00/0.00/0.00		0.00/0.00/0.00	-	-	DA Agg.	0.17/1.00/0.10	0.32/1.00/0.19	182			267.50				
	DA M3	0.50/1.00/0.33		0.50/1.00/0.33	-	-	DA M4	0.00/0.00/0.00	0.00/0.00/0.00	-			-				
	Qwen	8		LW-TD	1.00/1.00/1.00	1.00/1.00/1.00	73	180.61	Qwen	8			DA M2	0.18/0.50/0.11	0.20/1.00/0.11	-	-
LW-TB			1.00/1.00/1.00	1.00/1.00/1.00	35	105.20	DA M1	0.00/0.00/0.00			0.00/1.00/0.00	-	-				
LW-BD			1.00/1.00/1.00	1.00/1.00/1.00	41	114.59	LW-TD	0.67/1.00/0.50			0.67/1.00/0.50	33	67.10				
LW-BB			1.00/1.00/1.00	1.00/1.00/1.00	225	534.24	LW-TB	0.40/0.50/0.33			0.57/0.50/0.67	27	65.26				
LW Agg.			1.00/1.00/1.00	1.00/1.00/1.00	374	934.64	LW-BD	0.40/0.50/0.33			0.57/0.50/0.67	33	76.46				
LW M4			1.00/1.00/1.00	1.00/1.00/1.00	-	-	LW-BB	0.67/1.00/0.50			0.67/1.00/0.50	224	447.75				
DA-TD			1.00/1.00/1.00	1.00/1.00/1.00	9	24.91	LW Agg.	0.50/1.00/0.33			0.67/1.00/0.50	317	656.58				
DA-TB			1.00/1.00/1.00	1.00/1.00/1.00	9	25.84	LW M4	0.67/0.50/1.00			0.67/0.50/1.00	-	-				
DA-BD			1.00/1.00/1.00	1.00/1.00/1.00	22	32.26	LW M2	0.67/1.00/0.50			0.67/1.00/0.50	-	-				
DA-BB			1.00/1.00/1.00	1.00/1.00/1.00	22	33.06	DA-TD	0.29/0.50/0.20			0.55/0.50/0.60	23	75.25				
DA Agg.			1.00/1.00/1.00	1.00/1.00/1.00	62	116.08	DA-TB	0.29/0.50/0.20			0.55/0.50/0.60	25	79.16				
DA M4			1.00/1.00/1.00	1.00/1.00/1.00	-	-	DA-BD	0.67/1.00/0.50			0.67/1.00/0.50	32	67.44				
							DA-BB	0.67/1.00/0.50			0.67/1.00/0.50	39	79.30				
							DA Agg.	0.50/1.00/0.33			0.80/1.00/0.67	119	301.15				
					DA M4	0.40/0.50/0.33	0.40/0.50/0.33	-	-								
					DA M2	0.67/1.00/0.50	0.67/1.00/0.50	-	-								

foreign keys, functional dependencies, inclusion dependencies, join dependencies, order dependencies, etc. [54]. Models may include relational, bag, tree, graph, spatial, temporal and uncertain data.

There are also plenty of directions for information systems' research, which investigate *how well* humans use the technology.

Table 8: LLM-based interview results for tables in the Hockey database with *LLM-assisted prime filtering*.

Table Model	[T]	[P]	Prime F/R/P	Strat.	Exact F/R/P	Relaxed F/R/P	#Q	Time min	Table Model	[T]	[P]	Prime F/R/P	Strat.	Exact F/R/P	Relaxed F/R/P	#Q	Time min		
Scoring Shootout	DS	7	3	1.00/1.00/1.00	LW-TD	1.00/1.00/1.00	1.00/1.00/1.00	4	4.22	Goalies Shootout	DS	8	3	1.00/1.00/1.00	LW-TD	1.00/1.00/1.00	1.00/1.00/1.00	4	4.59
					LW-TB	1.00/1.00/1.00	1.00/1.00/1.00	4	4.22						LW-TB	1.00/1.00/1.00	1.00/1.00/1.00	4	4.59
					LW-BD	1.00/1.00/1.00	1.00/1.00/1.00	6	6.70						LW-BD	1.00/1.00/1.00	1.00/1.00/1.00	6	5.73
					LW-BB	1.00/1.00/1.00	1.00/1.00/1.00	8	8.81						LW-BB	1.00/1.00/1.00	1.00/1.00/1.00	8	7.53
					LW Agg.	1.00/1.00/1.00	1.00/1.00/1.00	22	23.94						LW Agg.	1.00/1.00/1.00	1.00/1.00/1.00	22	22.45
					LW M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-						LW M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-
					DA-TD	1.00/1.00/1.00	1.00/1.00/1.00	4	4.92						DA-TD	1.00/1.00/1.00	1.00/1.00/1.00	4	4.52
					DA-TB	1.00/1.00/1.00	1.00/1.00/1.00	4	4.64						DA-TB	1.00/1.00/1.00	1.00/1.00/1.00	4	4.51
					DA-BD	1.00/1.00/1.00	1.00/1.00/1.00	7	8.01						DA-BD	1.00/1.00/1.00	1.00/1.00/1.00	7	6.79
					DA-BB	1.00/1.00/1.00	1.00/1.00/1.00	7	8.01						DA-BB	1.00/1.00/1.00	1.00/1.00/1.00	7	6.79
DA Agg.	1.00/1.00/1.00	1.00/1.00/1.00	22	25.57	DA Agg.	1.00/1.00/1.00	1.00/1.00/1.00	22	22.61										
DA M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-	DA M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-										
Scoring Shootout	Qwen	7	3	1.00/1.00/1.00	LW-TD	1.00/1.00/1.00	1.00/1.00/1.00	4	5.12	Goalies Shootout	Qwen	8	3	1.00/1.00/1.00	LW-TD	1.00/1.00/1.00	1.00/1.00/1.00	4	4.47
					LW-TB	1.00/1.00/1.00	1.00/1.00/1.00	4	5.11						LW-TB	1.00/1.00/1.00	1.00/1.00/1.00	4	4.41
					LW-BD	1.00/1.00/1.00	1.00/1.00/1.00	6	7.46						LW-BD	1.00/1.00/1.00	1.00/1.00/1.00	6	7.47
					LW-BB	1.00/1.00/1.00	1.00/1.00/1.00	8	10.19						LW-BB	1.00/1.00/1.00	1.00/1.00/1.00	8	12.71
					LW Agg.	1.00/1.00/1.00	1.00/1.00/1.00	22	27.87						LW Agg.	1.00/1.00/1.00	1.00/1.00/1.00	22	29.11
					LW M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-						LW M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-
					DA-TD	1.00/1.00/1.00	1.00/1.00/1.00	4	5.53						DA-TD	1.00/1.00/1.00	1.00/1.00/1.00	4	6.09
					DA-TB	1.00/1.00/1.00	1.00/1.00/1.00	4	6.10						DA-TB	1.00/1.00/1.00	1.00/1.00/1.00	4	5.60
					DA-BD	1.00/1.00/1.00	1.00/1.00/1.00	7	10.17						DA-BD	1.00/1.00/1.00	1.00/1.00/1.00	7	12.50
					DA-BB	1.00/1.00/1.00	1.00/1.00/1.00	7	8.76						DA-BB	1.00/1.00/1.00	1.00/1.00/1.00	7	12.07
DA Agg.	1.00/1.00/1.00	1.00/1.00/1.00	22	30.56	DA Agg.	1.00/1.00/1.00	1.00/1.00/1.00	22	36.26										
DA M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-	DA M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-										
Team Vs Team	DS	8	4	0.86/1.00/0.75	LW-TD	1.00/1.00/1.00	1.00/1.00/1.00	8	9.13	Combined Shootouts	DS	8	5	1.00/1.00/1.00	LW-TD	0.00/0.00/0.00	0.00/1.00/0.00	15	17.94
					LW-TB	1.00/1.00/1.00	1.00/1.00/1.00	5	6.17						LW-TB	1.00/1.00/1.00	1.00/1.00/1.00	7	9.26
					LW-BD	1.00/1.00/1.00	1.00/1.00/1.00	8	9.13						LW-BD	1.00/1.00/1.00	1.00/1.00/1.00	11	13.88
					LW-BB	1.00/1.00/1.00	1.00/1.00/1.00	15	16.97						LW-BB	0.00/0.00/0.00	0.00/1.00/0.00	24	25.47
					LW Agg.	1.00/1.00/1.00	1.00/1.00/1.00	36	40.50						LW Agg.	0.67/1.00/0.50	0.67/1.00/0.50	57	66.56
					LW M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-						LW M2	0.00/0.00/0.00	0.00/1.00/0.00	-	-
					DA-TD	1.00/1.00/1.00	1.00/1.00/1.00	5	4.95						DA-TD	0.00/0.00/0.00	0.00/1.00/0.00	10	13.68
					DA-TB	1.00/1.00/1.00	1.00/1.00/1.00	5	4.95						DA-TB	0.00/0.00/0.00	0.00/1.00/0.00	10	15.22
					DA-BD	1.00/1.00/1.00	1.00/1.00/1.00	10	9.86						DA-BD	0.00/0.00/0.00	0.00/1.00/0.00	14	15.63
					DA-BB	1.00/1.00/1.00	1.00/1.00/1.00	10	9.86						DA-BB	0.00/0.00/0.00	0.00/1.00/0.00	15	15.96
DA Agg.	1.00/1.00/1.00	1.00/1.00/1.00	30	29.64	DA Agg.	0.00/0.00/0.00	0.00/1.00/0.00	49	60.49										
DA M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-	DA M4	0.00/0.00/0.00	0.00/0.50/0.00	-	-										
Team Vs Team	Qwen	8	4	0.86/1.00/0.75	LW-TD	1.00/1.00/1.00	1.00/1.00/1.00	5	7.55	Combined Shootouts	Qwen	8	5	1.00/1.00/1.00	DA M2	0.00/0.00/0.00	0.00/1.00/0.00	-	-
					LW-TB	1.00/1.00/1.00	1.00/1.00/1.00	5	7.56						LW-TD	0.80/1.00/0.67	0.80/1.00/0.67	13	19.12
					LW-BD	1.00/1.00/1.00	1.00/1.00/1.00	7	9.86						LW-TB	0.80/1.00/0.67	0.80/1.00/0.67	9	14.53
					LW-BB	1.00/1.00/1.00	1.00/1.00/1.00	15	22.84						LW-BD	1.00/1.00/1.00	1.00/1.00/1.00	10	14.10
					LW Agg.	1.00/1.00/1.00	1.00/1.00/1.00	32	47.80						LW-BB	0.80/1.00/0.67	0.80/1.00/0.67	31	47.69
					LW M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-						LW Agg.	0.80/1.00/0.67	0.80/1.00/0.67	63	95.45
					DA-TD	1.00/1.00/1.00	1.00/1.00/1.00	5	7.92						DA-TD	0.00/0.00/0.00	0.00/1.00/0.00	-	-
					DA-TB	1.00/1.00/1.00	1.00/1.00/1.00	5	7.82						LW M3	0.80/1.00/0.67	0.80/1.00/0.67	-	-
					DA-BD	1.00/1.00/1.00	1.00/1.00/1.00	10	17.32						DA-TD	0.00/0.00/0.00	0.00/0.00/1.00	6	11.04
					DA-BB	1.00/1.00/1.00	1.00/1.00/1.00	10	15.76						DA-TB	0.00/0.00/0.00	0.00/0.00/1.00	6	12.43
DA Agg.	1.00/1.00/1.00	1.00/1.00/1.00	30	48.82	DA-BD	0.67/0.50/1.00	0.67/0.50/1.00	15	24.09										
DA M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-	DA-BB	0.67/0.50/1.00	0.67/0.50/1.00	15	25.67										
Teams Half	DS	11	3	1.00/1.00/1.00	LW-TD	1.00/1.00/1.00	1.00/1.00/1.00	4	4.41	Teams Post	DS	17	3	0.80/1.00/0.67	DA Agg.	0.50/0.50/0.50	0.67/0.50/1.00	42	73.24
					LW-TB	1.00/1.00/1.00	1.00/1.00/1.00	4	4.41						DA M2	0.67/0.50/1.00	0.67/0.50/1.00	-	-
					LW-BD	1.00/1.00/1.00	1.00/1.00/1.00	6	6.25						LW-TD	0.67/1.00/0.50	0.67/1.00/0.50	5	4.90
					LW-BB	1.00/1.00/1.00	1.00/1.00/1.00	8	8.59						LW-TB	0.67/1.00/0.50	0.67/1.00/0.50	5	4.90
					LW Agg.	1.00/1.00/1.00	1.00/1.00/1.00	22	23.67						LW-BD	0.67/1.00/0.50	0.67/1.00/0.50	6	5.70
					LW M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-						LW-BB	0.67/1.00/0.50	0.67/1.00/0.50	7	6.63
					DA-TD	1.00/1.00/1.00	1.00/1.00/1.00	4	4.26						LW Agg.	0.67/1.00/0.50	0.67/1.00/0.50	23	22.13
					DA-TB	1.00/1.00/1.00	1.00/1.00/1.00	4	4.26						DA M4	0.67/1.00/0.50	0.67/1.00/0.50	-	-
					DA-BD	1.00/1.00/1.00	1.00/1.00/1.00	7	7.45						DA-TD	1.00/1.00/1.00	1.00/1.00/1.00	4	3.37
					DA-BB	1.00/1.00/1.00	1.00/1.00/1.00	7	7.45						DA-TB	1.00/1.00/1.00	1.00/1.00/1.00	4	3.36
DA Agg.	1.00/1.00/1.00	1.00/1.00/1.00	22	23.43	DA-BD	1.00/1.00/1.00	1.00/1.00/1.00	6	5.15										
DA M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-	DA-BB	1.00/1.00/1.00	1.00/1.00/1.00	6	5.15										
Teams Half	Qwen	11	3	1.00/1.00/1.00	LW-TD	1.00/1.00/1.00	1.00/1.00/1.00	4	6.00	Teams Post	Qwen	17	2	1.00/1.00/1.00	DA Agg.	1.00/1.00/1.00	1.00/1.00/1.00	20	17.03
					LW-TB	1.00/1.00/1.00	1.00/1.00/1.00	4	6.01						DA M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-
					LW-BD	1.00/1.00/1.00	1.00/1.00/1.00	6	8.19						LW-TD	1.00/1.00/1.00	1.00/1.00/1.00	3	5.11
					LW-BB	1.00/1.00/1.00	1.00/1.00/1.00	8	11.71						LW-TB	1.00/1.00/1.00	1.00/1.00/1.00	3	5.09
					LW Agg.	1.00/1.00/1.00	1.00/1.00/1.00	22	31.92						LW-BD	1.00/1.00/1.00	1.00/1.00/1.00	4	7.20
					LW M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-						LW-BB	1.00/1.00/1.00	1.00/1.00/1.00	4	7.31
					DA-TD	1.00/1.00/1.00	1.00/1.00/1.00	4	7.34						LW Agg.	1.00/1.00/1.00	1.00/1.00/1.00	14	24.72
					DA-TB	1.00/1.00/1.00	1.00/1.00/1.00	4	7.17						LW M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-
					DA-BD	1.00/1.00/1.00	1.00/1.00/1.00	7	11.77						DA-TD	1.00/1.00/1.00	1.00/1.00/1.00	3	5.72
					DA-BB	1.00/1.00/1.00	1.00/1.00/1.00	7	10.13						DA-TB	1.00/1.00/1.00	1.00/1.00/1.00	3	5.17
DA Agg.	1.00/1.00/1.00	1.00/1.00/1.00	22	36.42	DA-BD	1.00/1.00/1.00	1.00/1.00/1.00	4	8.15										
DA M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-	DA-BB	1.00/1.00/1.00	1.00/1.00/1.00	4	8.54										
Goalies	DS	23	4	0.86/1.00/0.75	LW-TD	0.67/1.00/0.50	0.67/1.00/0.50	8	12.26	Team Splits	DS	43	3	0.80/1.00/0.67	DA Agg.	1.00/1.00/1.00	1.00/1.00/1.00	14	27.57
					LW-TB	0.67/1.00/0.50	0.67/1.00/0.50	6	9.38						DA M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-
					LW-BD	0.67/1.00/0.50	0.67/1.00/0.50	9	12.70						LW-TD	1.00/1.00/1.00	1.00/1.00/1.00	4	3.42
					LW-BB	0.67/1.00/0.50	0.67/1.00/0.50	15	18.03						LW-TB	1.00/1.00/1.00	1.00/1.00/1.00	4	3.42
					LW Agg.	0.67/1.00/0.50	0.67/1.00/0.50	38	52.37						LW-BD	1.00/1.00/1.00	1.00/1.00/1.00	5	4.16
					LW M4	0.67/1.00/0.50	0.67/1.00/0.50	-	-						LW-BB	1.00/1.00/1.00	1.00/1.00/1.00	6	6.08
					DA-TD	0.67/1.00/0.50	0.67/1.00/0.50	9	10.90						LW Agg.	1.00/1.00/1.00	1.00/1.00/1.00	20	17.00
					DA-TB	0.67/1.00/0.50	0.67/1.00/0.50	8	9.75						LW M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-
					DA-BD	0.67/1.00/0.50	0.67/1.00/0.50	10	11.66						DA-TD	1.00/1.00/1.00	1.00/1.00/1.00	4	2.92
					DA-BB	0.67/1.00/0.50	0.67/1.00/0.50	11	11.65						DA-TB	1.00/1.00/1.00	1.00/1.00/1.00	4	2.92
DA Agg.	0.67/1.00/0.50	0.67/1.00/0.50	37	43.95	DA-BD	1.00/1.00/1.00	1.00/1.00/1.00	6	4.78										
DA M4	0.67/1.00/0.50	0.67/1.00/0.50	-	-	DA-BB	1.00/1.00/1.00	1.00/1.00/1.00	6	4.78										
Goalies	Qwen	23	3	1.00/1.00/1.00	LW-TD	1.00/1.00/1.00	1.00/1.00/1.00	4	5.49	Team Splits	Qwen	43	2	1.00/1.00/1.00	DA Agg.	1.00/1.00/1.00	1.00/1.00/1.00	14	31.41
					LW-TB	1.00/1.00/1.00	1.00/1.00/1.00	4	5.47						DA M4	1.00/1.00/1.00	1.00/1.00/1.00	-	-
					LW-BD	1.00/1.00/1.00	1.00/1.00/1.00	6	7.90						LW-TD	1.00/1.00/1.00	1.00/1.00/1.00	3	7.25
					LW-BB	1.00/1.00/1.00	1.00/1.00/1.00	8	10.21						LW-TB	1.00/1.00/1.00	1.00/1.00/1.00	4	7.28
					LW Agg.	1.00/1.00/1.00	1.00/1.00/1.00	22	29.06						LW-BD	1.00/1.00/1.00	1.00/1.00/1.00	4	8.00
					LW M4	1													

Table 9: LLM-based interview results for tables without ground-truth keys, ordered by schema size. Every row averages the interview over the prime-attribute budgets $|P| = 1, \dots, 5$, where $|P|^*$ is the predicted budget of the model. The four right-hand column pairs report the dualization-based family on the same interviews.

Table	T	Model	$ P ^*$	LW-TD		LW-TB		LW-BD		LW-BB		DA-TD		DA-TB		DA-BD		DA-BB	
				#Q	min	#Q	min	#Q	min	#Q	min	#Q	min	#Q	min	#Q	min	#Q	min
abalone	9	DS	1	2.2	2.33	2.0	2.17	5.2	4.62	10.4	8.69	4.0	4.12	4.0	4.12	7.2	6.09	7.2	6.09
		Qwen	1	3.8	4.98	3.8	4.97	6.0	6.77	12.4	14.02	3.8	4.41	3.8	4.39	8.0	7.99	8.0	7.97
routes	9	DS	5	7.6	8.13	6.4	6.70	8.4	8.37	12.4	12.46	5.8	5.45	5.6	5.18	9.2	8.38	9.4	8.59
		Qwen	5	5.6	7.36	5.0	6.57	6.6	7.88	11.4	13.66	4.8	6.22	5.0	6.56	8.0	9.24	8.8	10.25
breast	11	DS	1	5.6	4.61	7.0	5.63	5.0	4.13	7.2	5.91	4.0	3.24	4.0	3.24	4.0	3.22	4.0	3.22
		Qwen	1	4.6	4.79	5.6	6.57	4.8	4.51	7.2	6.80	4.0	4.21	4.0	4.18	4.0	3.43	4.0	3.43
echo	13	DS	1	5.6	6.30	6.4	6.90	5.0	5.10	7.2	7.45	5.0	4.95	5.0	4.95	5.8	6.26	5.2	5.23
		Qwen	1	3.0	4.66	3.0	4.66	5.2	7.86	11.8	18.65	4.6	8.08	4.6	8.13	7.6	11.96	8.2	12.77
bridges	13	DS	1	6.6	5.22	7.6	6.04	5.4	4.41	5.6	4.34	5.4	4.48	5.4	4.48	4.4	3.62	4.4	3.62
		Qwen	1	8.2	10.86	7.0	10.27	8.4	11.95	7.2	8.29	4.0	5.69	4.0	5.68	4.0	4.75	4.0	4.77
claims	13	DS	1	6.6	5.33	7.8	6.70	6.0	4.91	7.2	5.78	4.0	3.19	4.0	3.19	4.0	2.99	4.0	2.99
		Qwen	1	6.4	7.87	6.8	9.69	5.6	5.71	7.2	6.17	4.0	4.59	4.0	4.57	4.0	3.48	4.0	3.48
pdbx	13	DS	2	7.8	8.42	5.8	6.06	7.8	7.98	9.6	10.03	5.2	5.57	5.4	5.63	7.6	8.14	8.0	8.55
		Qwen	2	7.8	9.09	4.8	5.54	6.4	6.91	10.2	10.81	4.0	4.11	4.0	4.12	6.0	5.72	6.0	5.74
adult	15	DS	1	3.4	2.86	3.4	2.86	4.6	3.65	6.2	4.93	3.2	3.11	3.2	3.11	5.0	4.15	5.0	4.16
		Qwen	1	3.8	2.91	3.8	2.90	6.0	4.21	12.4	8.34	3.8	2.84	3.8	2.86	8.0	5.63	8.0	5.65
hospital	15	DS	1	7.0	6.24	10.6	9.80	7.2	6.41	5.2	4.56	8.4	7.36	9.0	7.82	5.4	4.89	5.4	4.89
		Qwen	1	9.0	12.47	7.0	9.63	7.8	10.57	7.2	9.54	4.0	4.96	4.0	4.97	4.0	4.46	4.0	4.47
lineitem	16	DS	2	8.6	8.66	5.2	5.22	7.4	7.91	9.8	9.62	5.2	5.11	5.2	5.11	7.0	6.64	7.4	7.05
		Qwen	2	5.0	7.06	4.8	6.92	5.6	7.23	10.2	13.67	4.4	5.70	4.4	5.67	6.6	8.31	6.6	8.28
weather	18	DS	3	3.4	2.75	3.4	2.75	7.4	6.42	12.4	15.51	6.0	5.91	5.8	5.67	8.4	7.58	8.6	7.93
		Qwen	3	5.4	8.37	4.0	6.14	6.4	9.11	11.6	17.05	4.8	7.08	4.8	7.05	7.2	9.56	7.2	9.57
ncvoter	19	DS	2	5.8	4.46	8.4	6.63	5.2	4.24	5.2	3.98	7.0	5.70	7.0	5.70	4.8	3.84	4.8	3.84
		Qwen	2	4.6	6.63	9.8	17.03	4.8	6.11	5.0	5.46	6.6	8.68	6.6	8.75	4.6	5.90	4.6	5.86